

Transport Layer Services

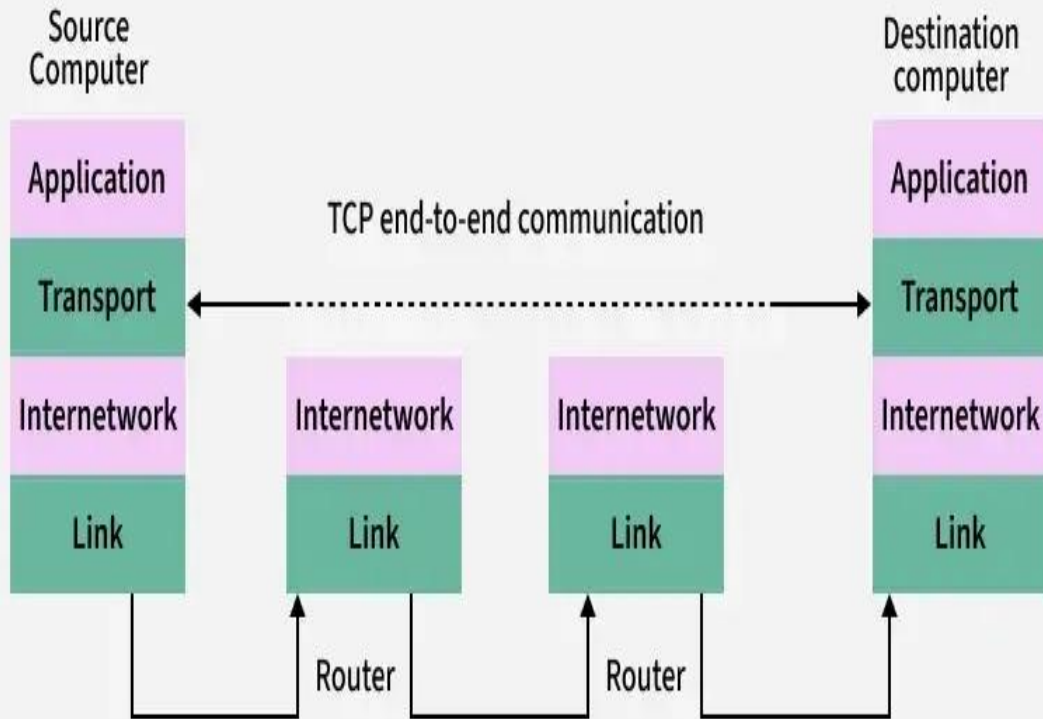
Services Offered by Transport Layer

The transport layer provides reliable data transfer services such as segmentation, flow control, error detection, and retransmission, end-to-end communication between devices.

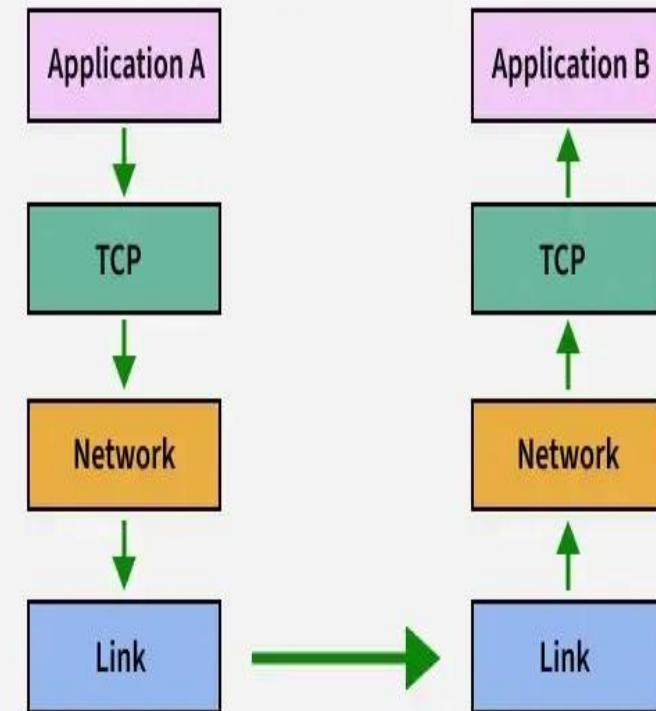


Services Offered by Transport Layer

1. End-to-End communication



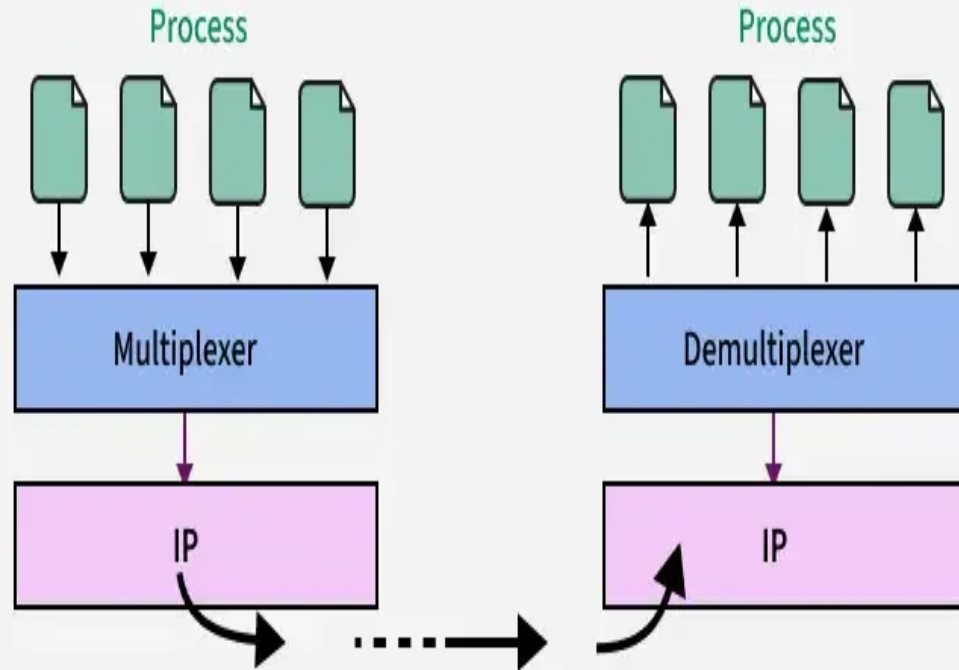
2. Flow Control



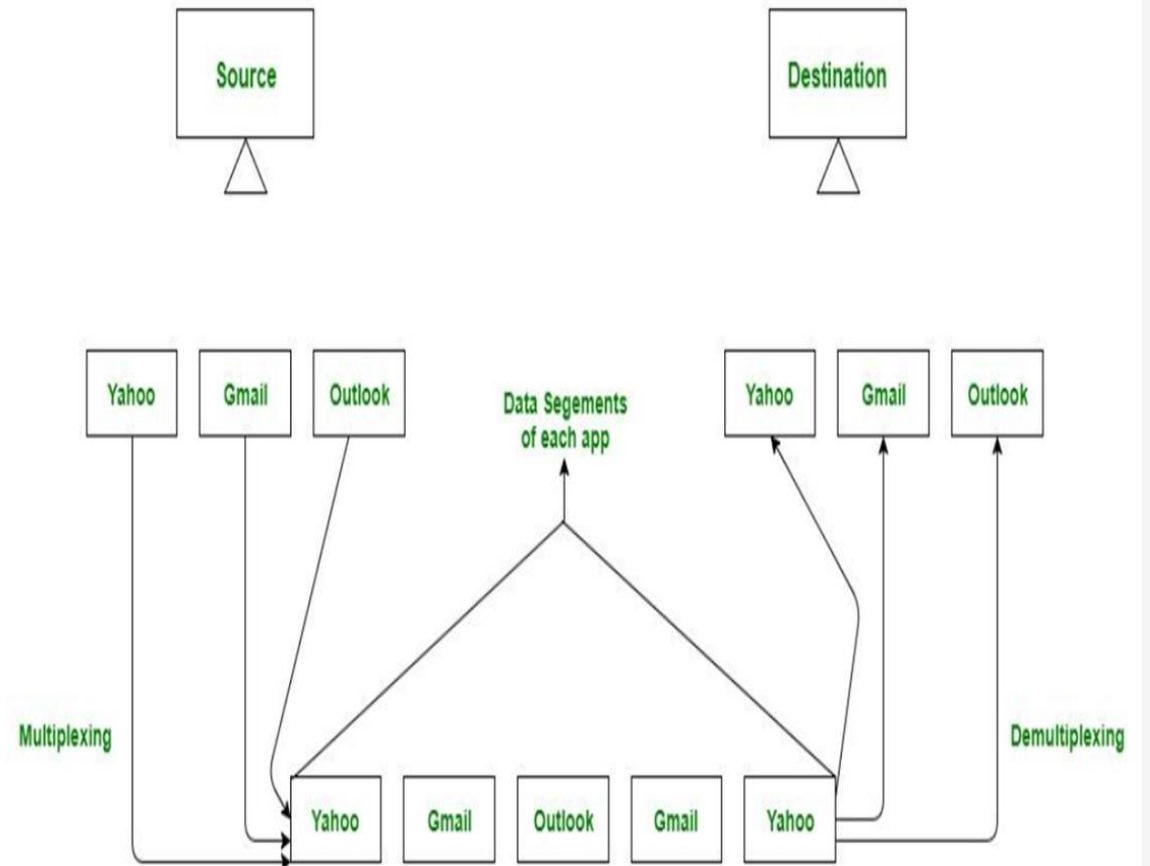
Services Offered by Transport Layer

3. Multiplexing and Demultiplexing

It manages multiple data streams on the same network connection by assigning unique ports.

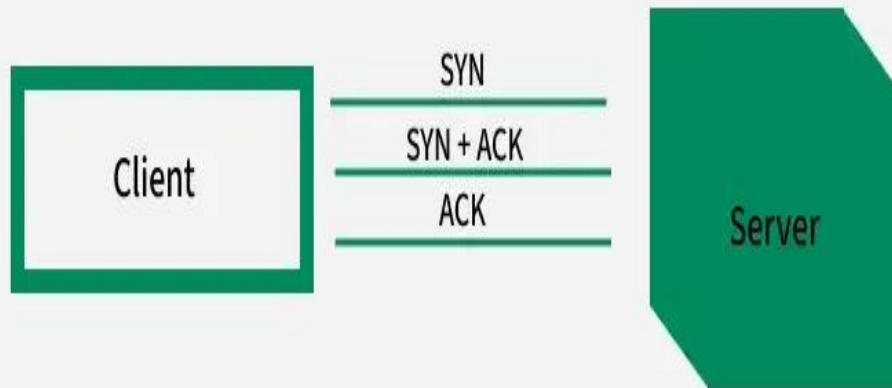


Multiplexing / Demultiplexing



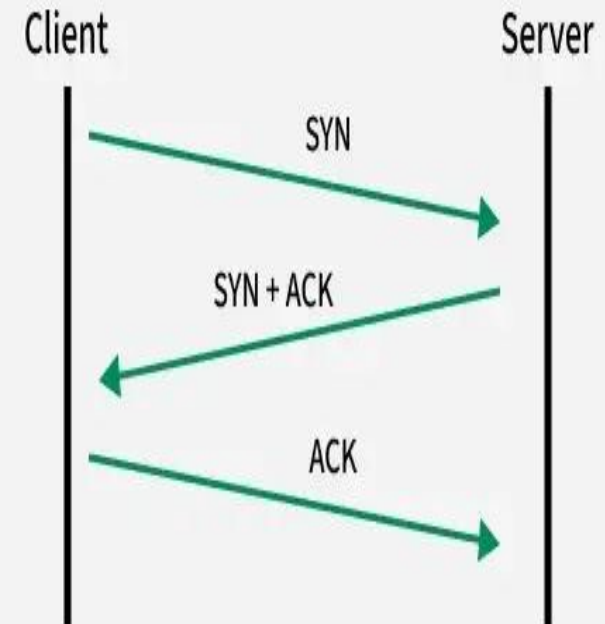
Services Offered by Transport Layer

4. Connection Establishment



TCP 3 Way Handshake

5. Connection Termination



Services Offered by Transport Layer

6. Reliable Data Delivery

It ensures all data is delivered correctly, in sequence, and without duplication (in protocols like TCP).

7. Quality of Service (QoS) (Optional)

It ensures priority and performance for specific types of data, like video calls or file transfers.

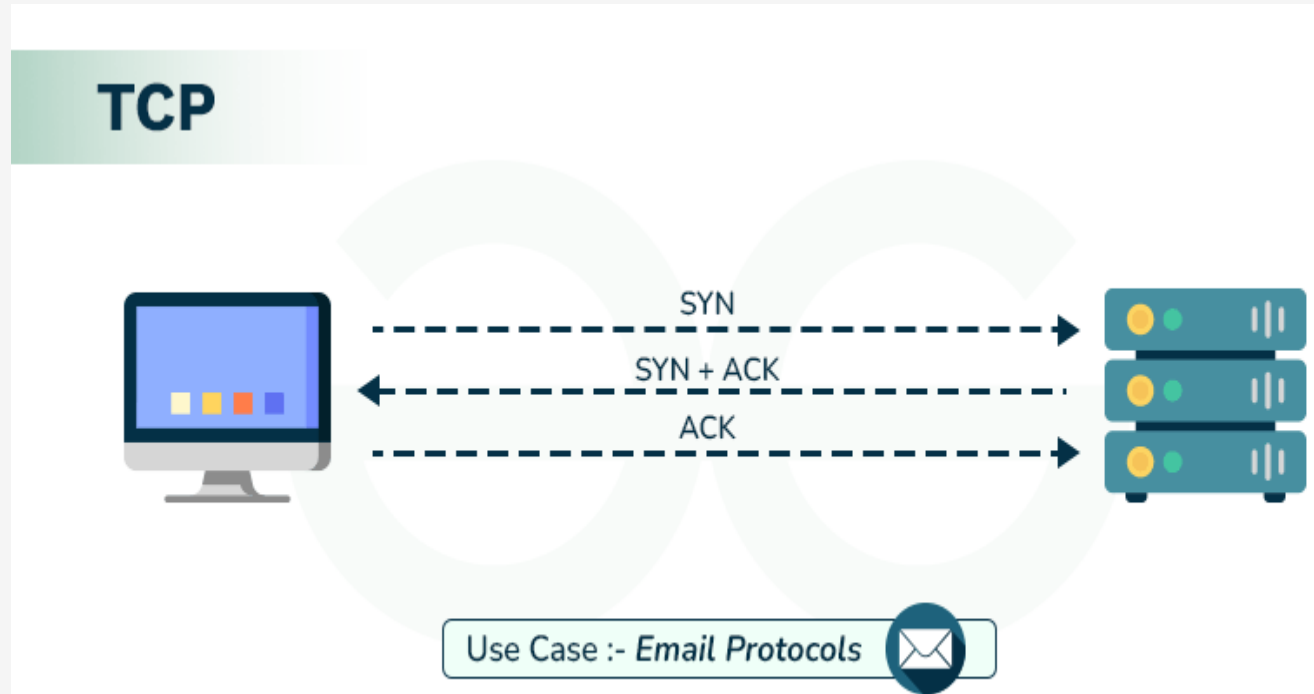
TCP Connection Establishment

Real-Life Analogy: Making a Phone Call

Imagine you want to call a friend:

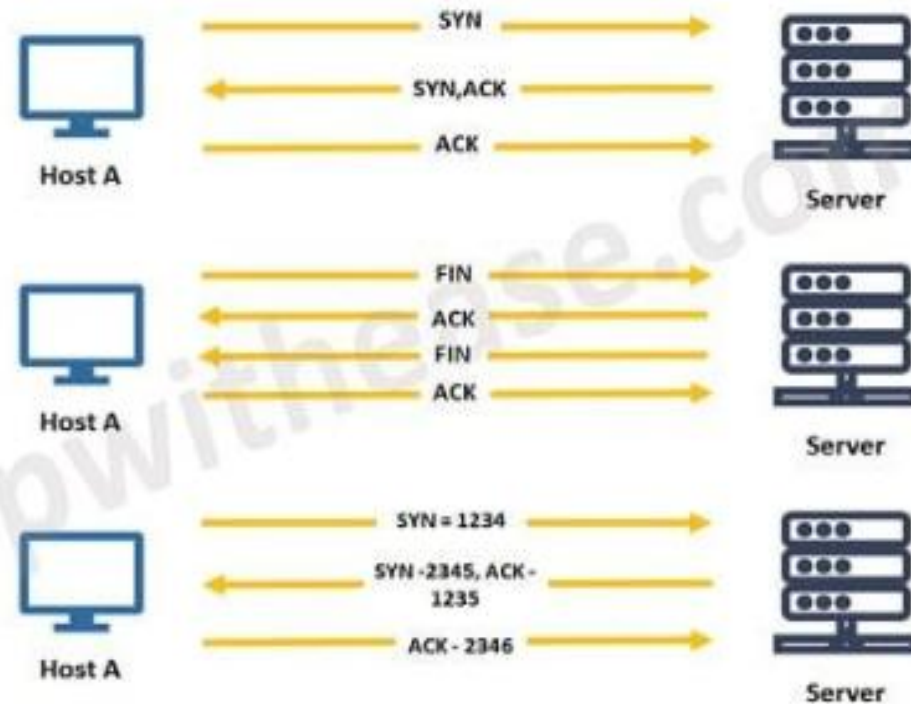
- You dial their number – this is like saying “Hey, I want to talk!”
- Your friend picks up and says, “Hello!” – they’re acknowledging your request.
- You say, “Great, let’s talk!” – now both of you are ready to chat.

This back-and-forth ensures both sides are ready and listening before the actual conversation begins.



TCP Three-Way Handshake (Step-by-Step)

TCP 3-Way Handshake



The three-way handshake ensures that both parties are synchronized and ready for reliable communication. This process helps prevent data loss, ensures data integrity, and sets the foundation for a secure and orderly data transfer.

TCP connection establishment is a crucial aspect of networking, providing a robust mechanism for devices to establish connections before engaging in data exchange. Understanding this process is essential for network administrators, developers, and anyone involved in building or maintaining networked systems.

Connection Teardown

The process of terminating a TCP connection typically involves a four-step process known as the four-way handshake:

FIN (Finish): One side (usually the client) sends a FIN packet to indicate that it has finished sending data. This packet also includes the sequence number.

ACK (Acknowledge): The receiving side acknowledges the FIN packet by sending an ACK packet back, confirming that it has received the FIN.

FIN (Finish): The receiving side then sends its own FIN packet to indicate that it is also finished sending data.

ACK (Acknowledge): Finally, the original sender acknowledges the receipt of the FIN packet with an ACK, completing the connection teardown process.

This structured approach to connection teardown ensures that all data is transmitted and acknowledged before the connection is fully closed, preventing data loss and ensuring a clean termination of the session.

TCP Connection Establishment: Pros & Cons

Advantages (Importance)

Reliability & Data Integrity: Ensures error-checked channel and correct sequence/order.

Orderly Data Transfer: Sets up an organized process.

Error Detection and Recovery: Mechanisms to manage connection errors.

Flow Control: Prevents the sender from overwhelming the receiver.

Compatibility & Security: Allows seamless operation with firewalls and enables features like SSL/TLS for encryption.

Disadvantages (Three-way Handshake)

Overhead: Requires exchanging multiple packets (SYN, SYN-ACK, ACK) *before* data transfer begins.

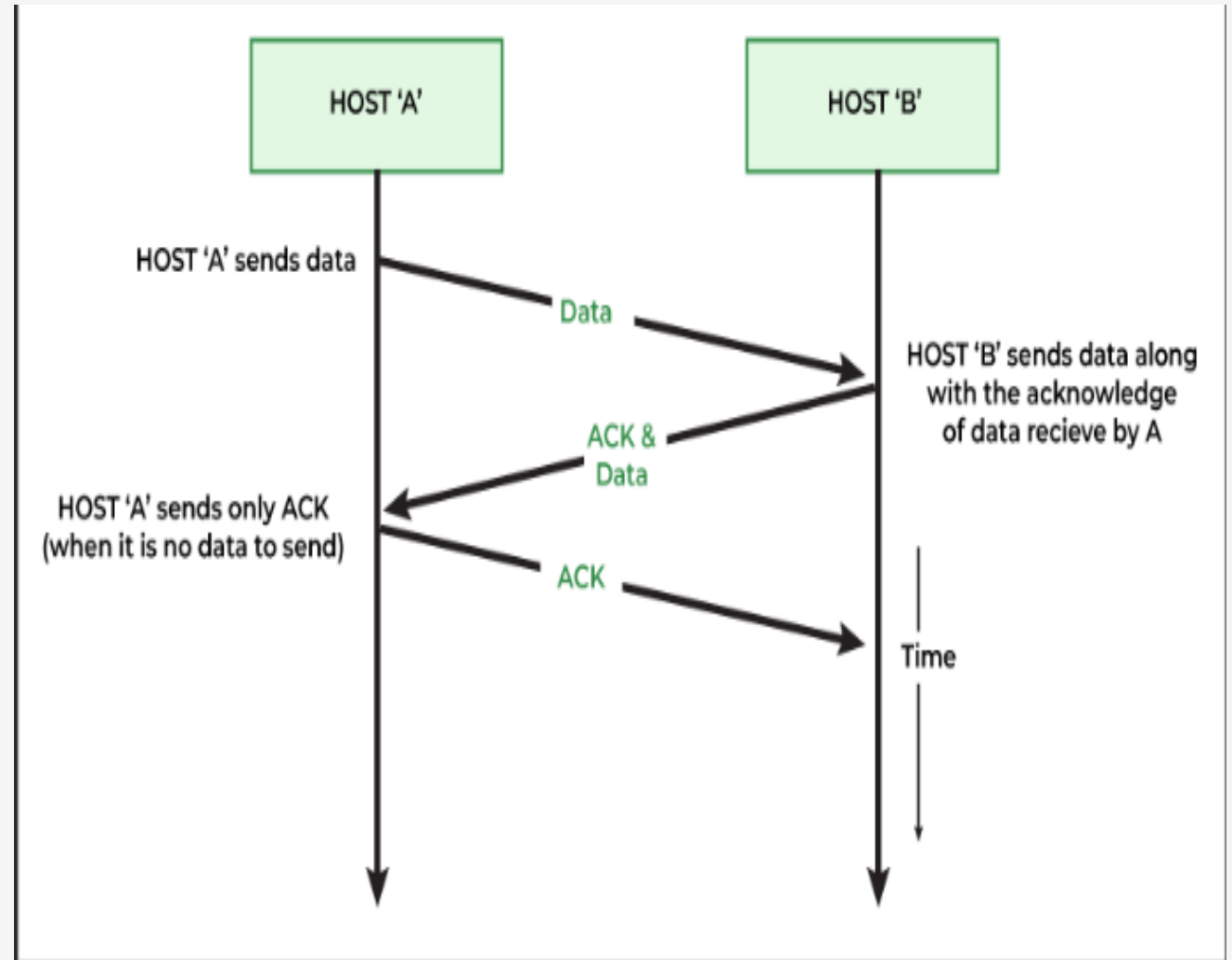
Connection Latency: Time taken for the handshake introduces a delay, problematic for quick connections (e.g., real-time apps).

Resource Utilization: The process consumes network resources, potentially impacting performance in high-traffic situations.

Unsuitability for Stateless Protocols: TCP's connection-oriented nature and required handshake are ill-suited for stateless protocols or apps where establishing and tearing down connections is unnecessary (e.g., protocols typically using UDP).

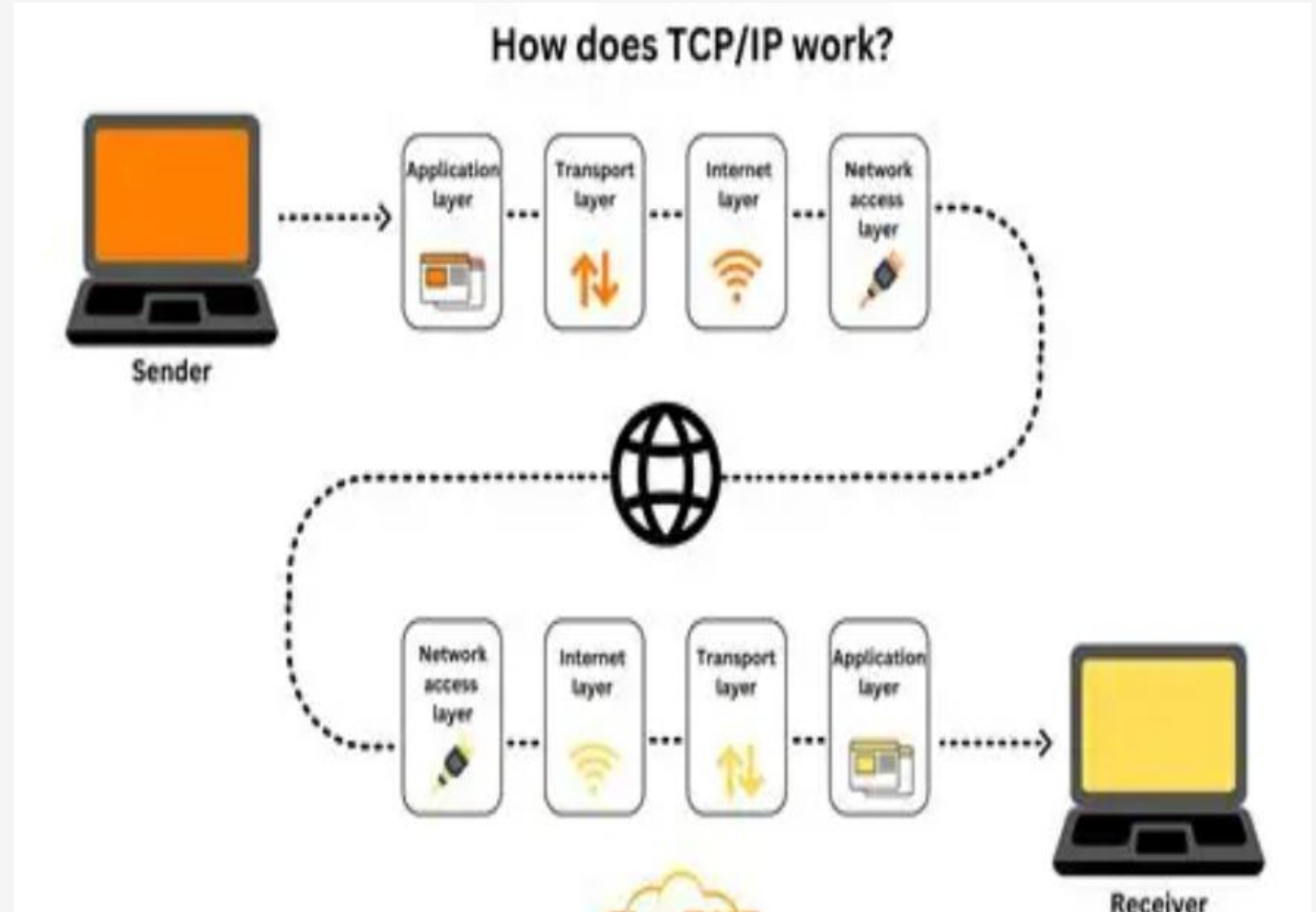
"Piggybacking"

in computer networks is a technique used in full-duplex communication to improve network efficiency by combining control information, typically an acknowledgment (ACK), with an outgoing data frame. Instead of sending a separate, small ACK packet, the receiver delays the acknowledgment slightly and attaches it to the next data packet it needs to send back to the original sender.



Conclusion

In conclusion, TCP connection establishment is the cornerstone of reliable and orderly communication over the internet. The meticulous three-way handshake, combined with other protocols and mechanisms, guarantees that data is exchanged securely and without loss. Understanding the nuances of TCP connection establishment empowers network administrators, developers, and enthusiasts to optimize their systems for peak performance. As technology continues to advance, the principles of TCP connection establishment will remain a fundamental element in the foundation of a connected world.



Dynamic Host Configuration Protocol (DHCP)

Is a network management protocol used to automate the process of assigning IP addresses and other communication parameters to devices on a network. This protocol eliminates the need for manual configuration of each device, making it essential for both small and large networks.

DHCP Packet Format

Operation Code	Hardware Type	Hardware Length	Hop Count
Transition ID			
Number of Seconds		Flags	
Client IP Address			
Your IP Address			
Server IP Address			
Gateway IP Address			
Client Hardware Address (16 Byte)			
Server Name (64 Byte)			
Boot File Name (128 Byte)			
Option (Variable Length)			

Components of DHCP



DHCP Server



DHCP Relay



DHCP Client



IP Address Pool



Subnets



Lease



DNS Servers

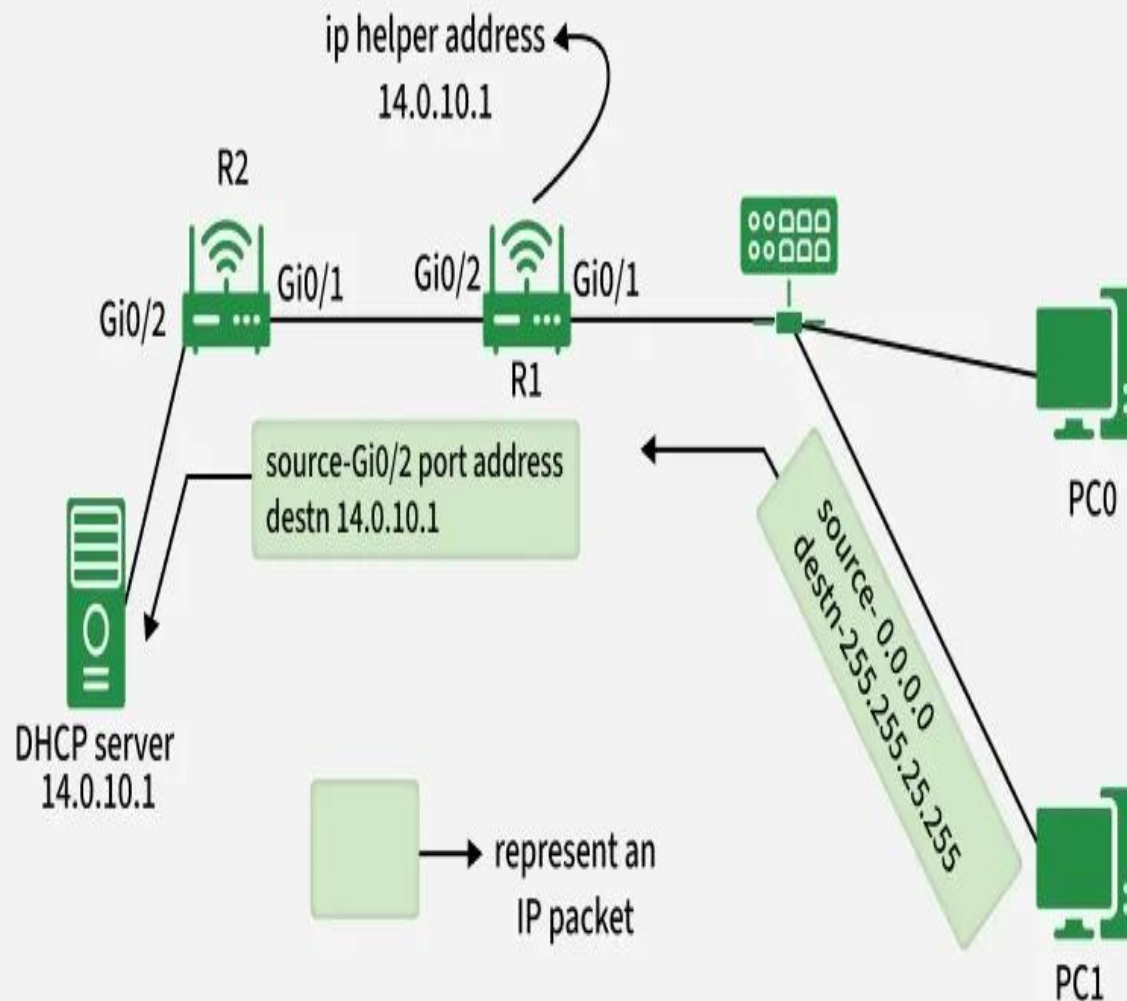


Default Gateway



Options

Working of DHCP



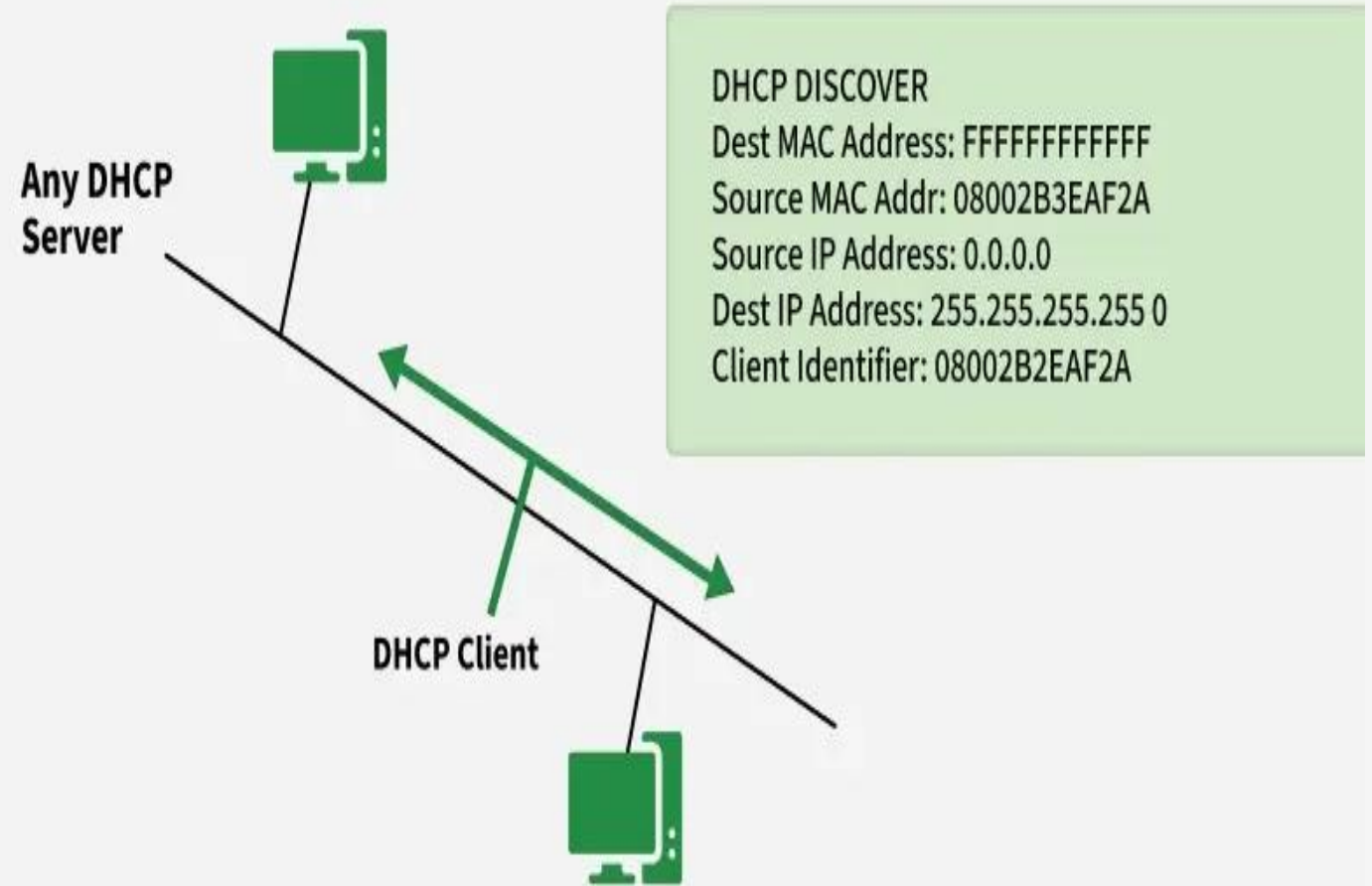
1. DHCP Discover Message:

- This is the first message generated in the communication process between the server and the client.
- This message is generated by the Client host in order to discover if there is any DHCP server/servers are present in a network or not.
- This message is broadcasted to all devices present in a network to find the DHCP server. This message is 342 or 576 bytes long.

. DHCP Discover Message:

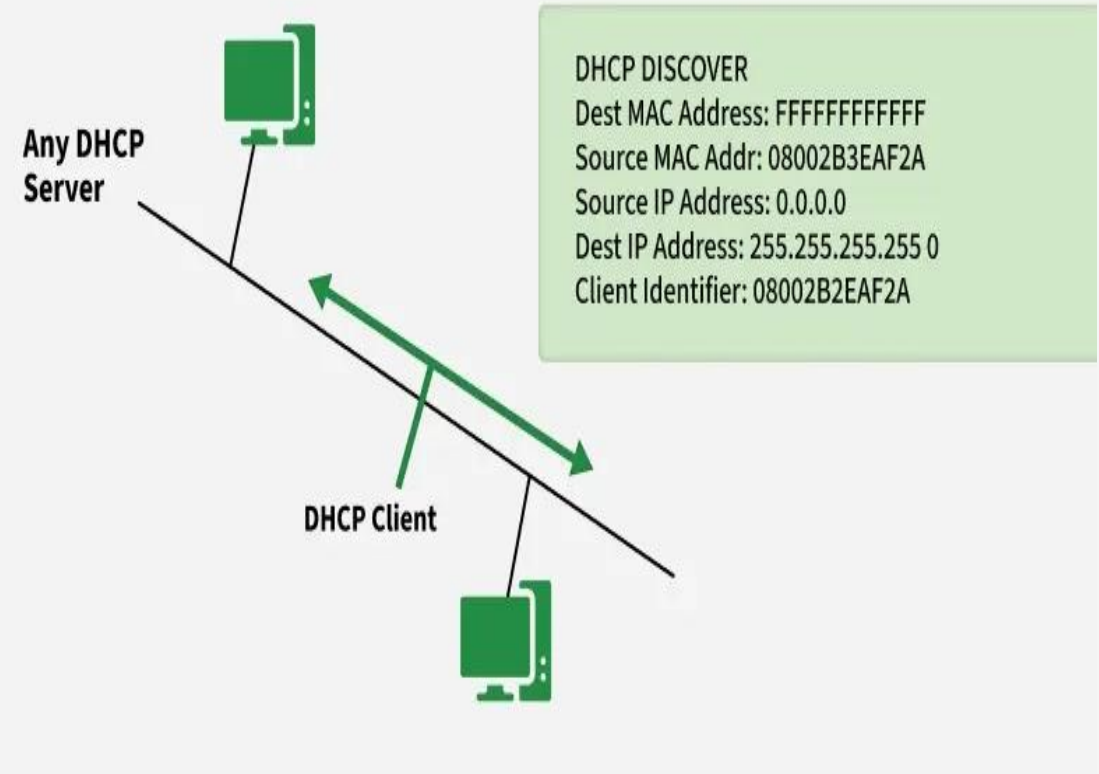
As shown in the figure, the source MAC address (client PC) is 08002B2EAF2A, the destination MAC address(server) is FFFFFFFF, the source IP address is 0.0.0.0(because the PC has had no IP address till now) and the destination IP address is 255.255.255.255 (IP address used for broadcasting).

As they discover message is broadcast to find out the DHCP server or servers in the network therefore broadcast IP address and MAC address is used.



DHCP Offers A Message

- Now, for the offer message, the source IP address is 172.16.32.12 (server's IP address in the example), the destination IP address is 255.255.255.255 (broadcast IP address).
- The source MAC address is 00AA00123456, the destination MAC address is 00:11:22:33:44:55 (client's MAC address).
- Here, the offer message is broadcast by the DHCP server therefore destination IP address is the broadcast IP address and destination MAC address is 00:11:22:33:44:55 (client's MAC address) and the source IP address is the server IP address and the MAC address is the server MAC address.



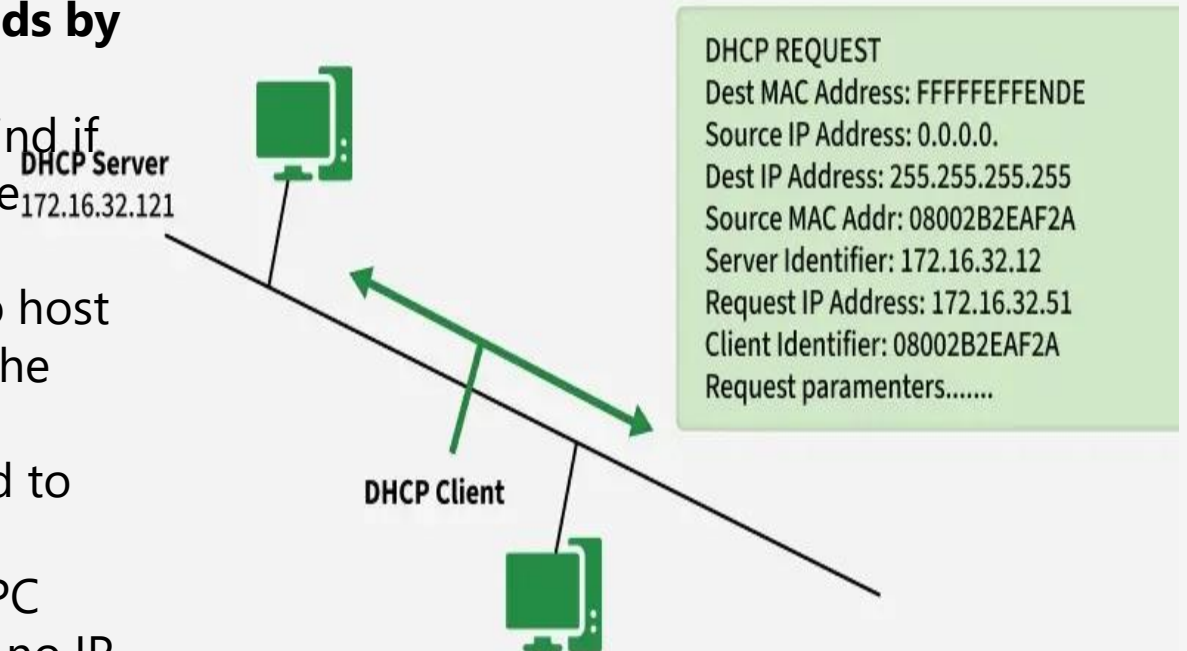
DHCP Request Message

When a client receives an offer message, it responds by broadcasting a DHCP request message.

The client will produce a gratuitous ARP in order to find if there is any other host present in the network with the same IP address.

If there is no reply from another host, then there is no host with the same TCP configuration in the network and the message is broadcasted to the server showing the acceptance of the IP address. A Client ID is also added to this message.

Now, the request message is broadcast by the client PC therefore source IP address is 0.0.0.0(as the client has no IP right now) and destination IP address is 255.255.255.255 (the broadcast IP address) and the source MAC address is 08002B2EAF2A (PC MAC address) and destination MAC address is FFFFFFFF.FFFF.

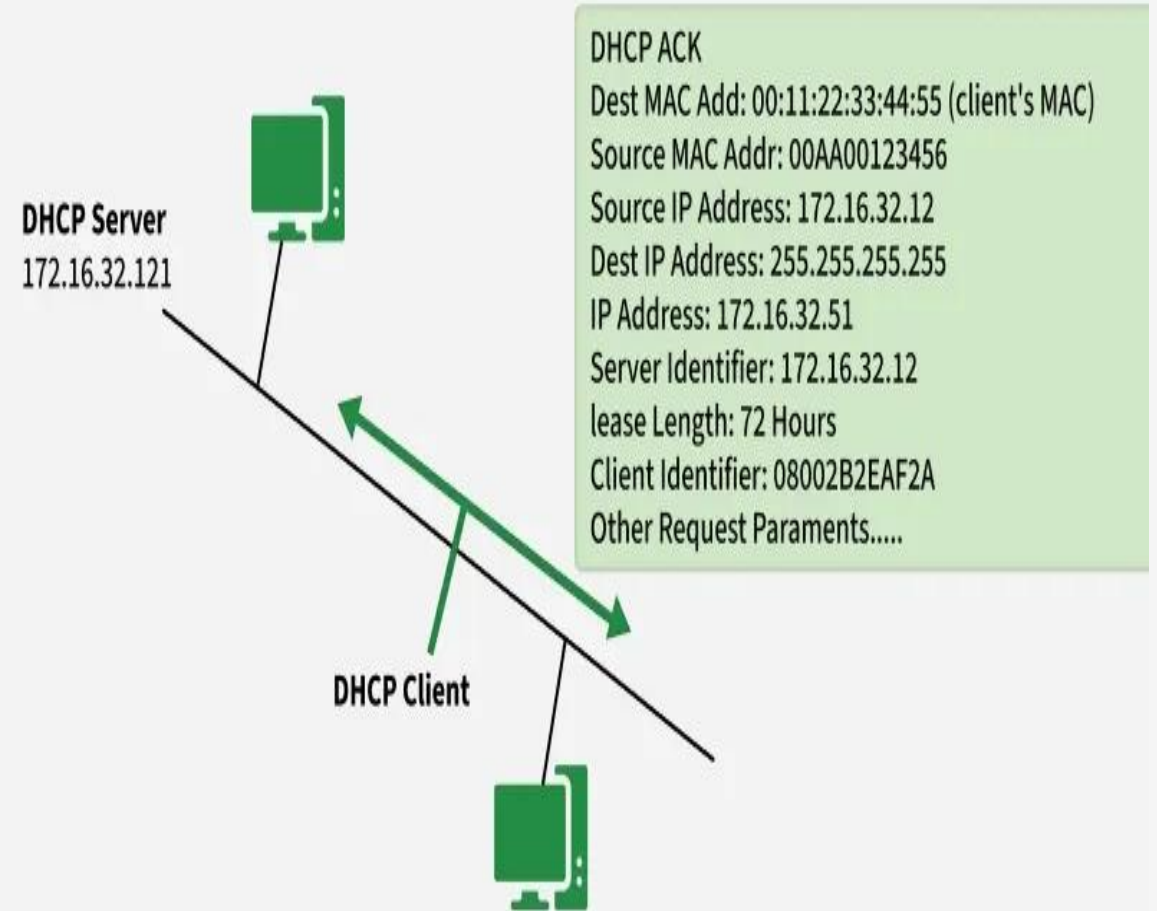


DHCP Acknowledgment Message

In response to the request message received, the server will make an entry with a specified client ID and bind the IP address offered with lease time. Now, the client will have the IP address provided by the server. Now the server will make an entry of the client host with the offered IP address and lease time.

This IP address will not be provided by the server to any other host.

The destination MAC address is 00:11:22:33:44:55 (client's MAC address) and the destination IP address is 255.255.255.255 and the source IP address is 172.16.32.12 and the source MAC address is 00AA00123456 (server MAC address).



DHCP Additional Messages

DHCP Message	Purpose
DHCP NACK	Sent by the server when a client requests an invalid IP address (e.g., if the IP pool is empty).
DHCP Decline	Sent by the client to the server if the offered IP address is already in use (e.g., detected via a Gratuitous ARP reply).
DHCP Release	Sent by the client to release an IP address back to the pool and cancel the remaining lease time.
DHCP Inform	Used by a client with a manually configured IP address to obtain <i>other</i> local network configuration parameters (like DNS or domain name) without requesting a new IP.

DORA is an acronym for the **four key steps** in the **Dynamic Host Configuration Protocol (DHCP)**, which automatically assigns IP addresses to devices on a network:

1. **Discover**

- The client sends a **DHCP Discover** message as a broadcast to locate available DHCP servers.
- This message includes the client's MAC address and a request for configuration parameters.

2. **Offer**

- DHCP servers respond with a **DHCP Offer** message.
- This includes an available IP address, subnet mask, lease duration, and other configuration details.

3. **Request**

- The client selects one of the offers and sends a **DHCP Request** message to the chosen server.

4. Acknowledge

- The server replies with a **DHCP Acknowledge (ACK)** message.
 - This finalizes the IP address assignment and completes the configuration.
-

Summary of Message Flow

Step	Message Type	Direction	Purpose
Discover	DHCP Discover	Client → Server	Find available DHCP servers
Offer	DHCP Offer	Server → Client	Propose IP address and config
Request	DHCP Request	Client → Server	Accept one offer
Acknowledge	DHCP ACK	Server → Client	Confirm assignment and lease



Key Notes

- All DHCP messages are typically **broadcast** at the network layer.
- The process ensures **automatic IP assignment**, reducing manual configuration.
- DORA is essential for **plug-and-play networking**, especially in dynamic environments like home Wi-Fi or enterprise LANs.

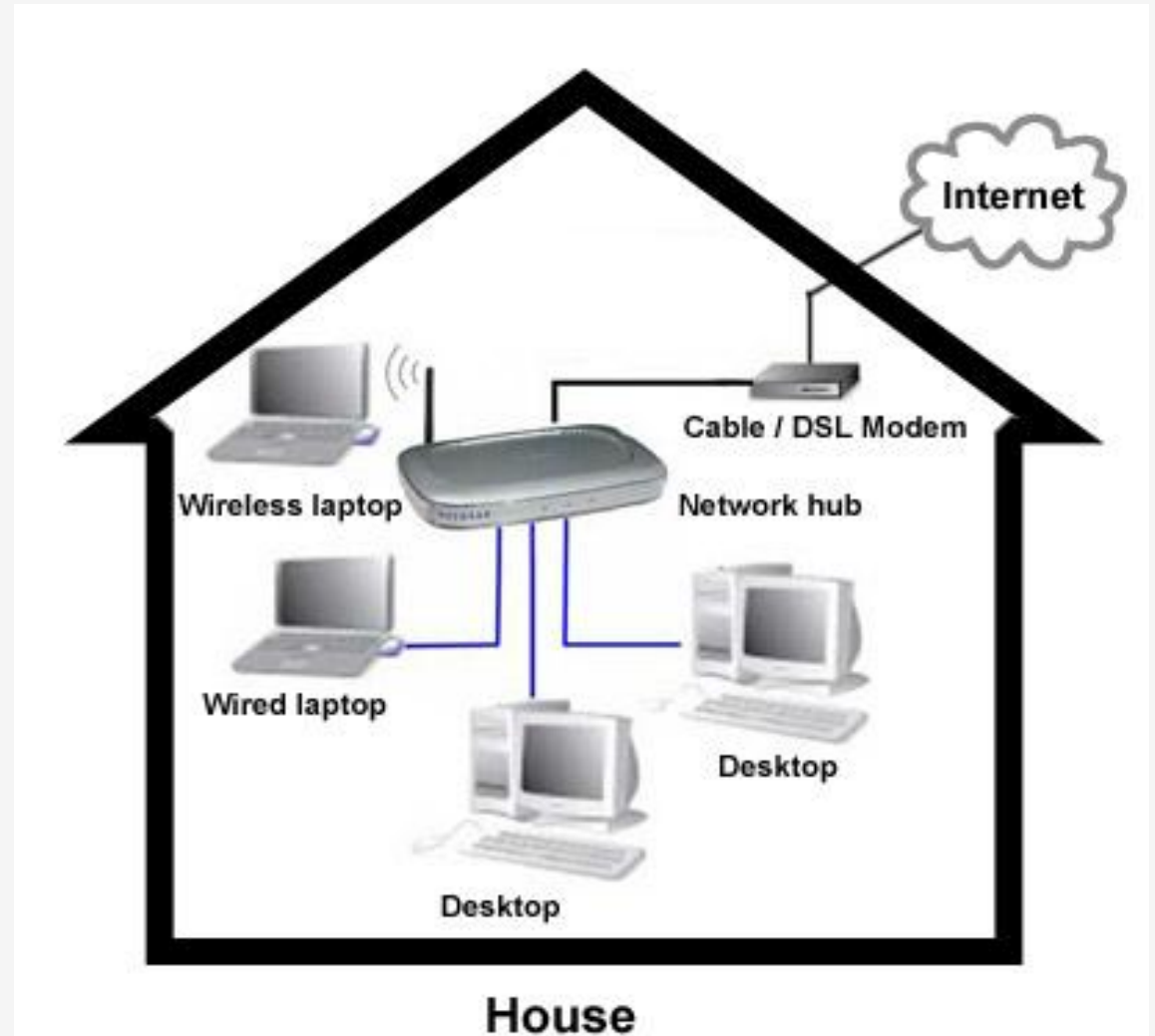
A **home network** connects all your devices—such as desktop computers, laptops, printers, smartphones, gaming consoles, and smart home devices—to the internet and to each other.

To set up a home network, you typically need:

A modem: This connects to your internet service provider.

A router: This device controls the home network and allows multiple devices to connect to the internet.

Devices: All your connected devices communicate through the router.



Wired networks

Wired connections rely on Ethernet cables to connect your devices directly to your network router, offering stability and speed ideal for high-bandwidth activities, like gaming or streaming in high definition.

Wireless networks (Wi-Fi)

Most common in homes, Wi-Fi networks offer the convenience of connecting devices without cables. This network type requires a wireless router and is flexible, but it can be affected by distance, physical obstructions, and interference.

Hybrid networks

Combining wired and wireless elements, hybrid networks can optimize the strengths of both types, providing reliability where needed while extending wireless convenience to portable devices.

Domain Name System (DNS)

DNS is a hierarchical and distributed naming system that translates domain names into IP addresses.

Step 01: User requests and local cache is checked



Computer browser requests to visit <https://geeksforgeeks.org>



Local Cache is checked for requested address

Step 02: Host Files are checked



If IP address not found in cache



Hosts File

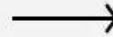


The hosts file is a system file that maps domain names to IP addresses locally

Step 03: Query DNS Resolver

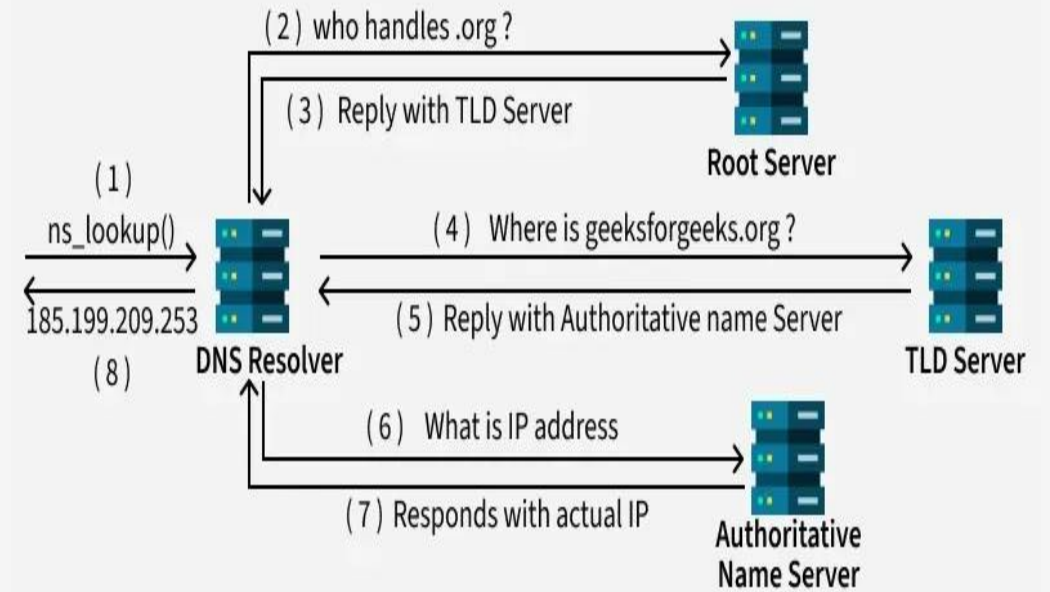


If IP address not found locally



DNS Resolver is a server provided by ISP

Step 04: DNS Search by DNS Resolver



Step 05: Return IP address to Computer



DNS Resolver

Return IP address to computer →



Computer

•**User Input:** You enter a website address (for example, www.geeksforgeeks.org) into your web browser.

•**Local Cache Check:** Your browser first checks its local cache to see if it has recently looked up the domain. If it finds the corresponding IP address, it uses that directly without querying external servers.

•**DNS Resolver Query:** If the IP address isn't in the local cache, your computer sends a request to a DNS resolver. The resolver is typically provided by your Internet Service Provider (ISP) or your network settings.

•**Root DNS Server:** The resolver sends the request to a root [DNS server](#). The root server doesn't know the exact IP address for www.geeksforgeeks.org but knows which Top-Level Domain (TLD) server to query based on the domain's extension (e.g., .org).

•**TLD Server:** The TLD server for .org directs the resolver to the authoritative DNS server for geeksforgeeks.org.

•**Authoritative DNS Server:** This server holds the actual DNS records for geeksforgeeks.org, including the IP address of the website's server. It sends this IP address back to the resolver.

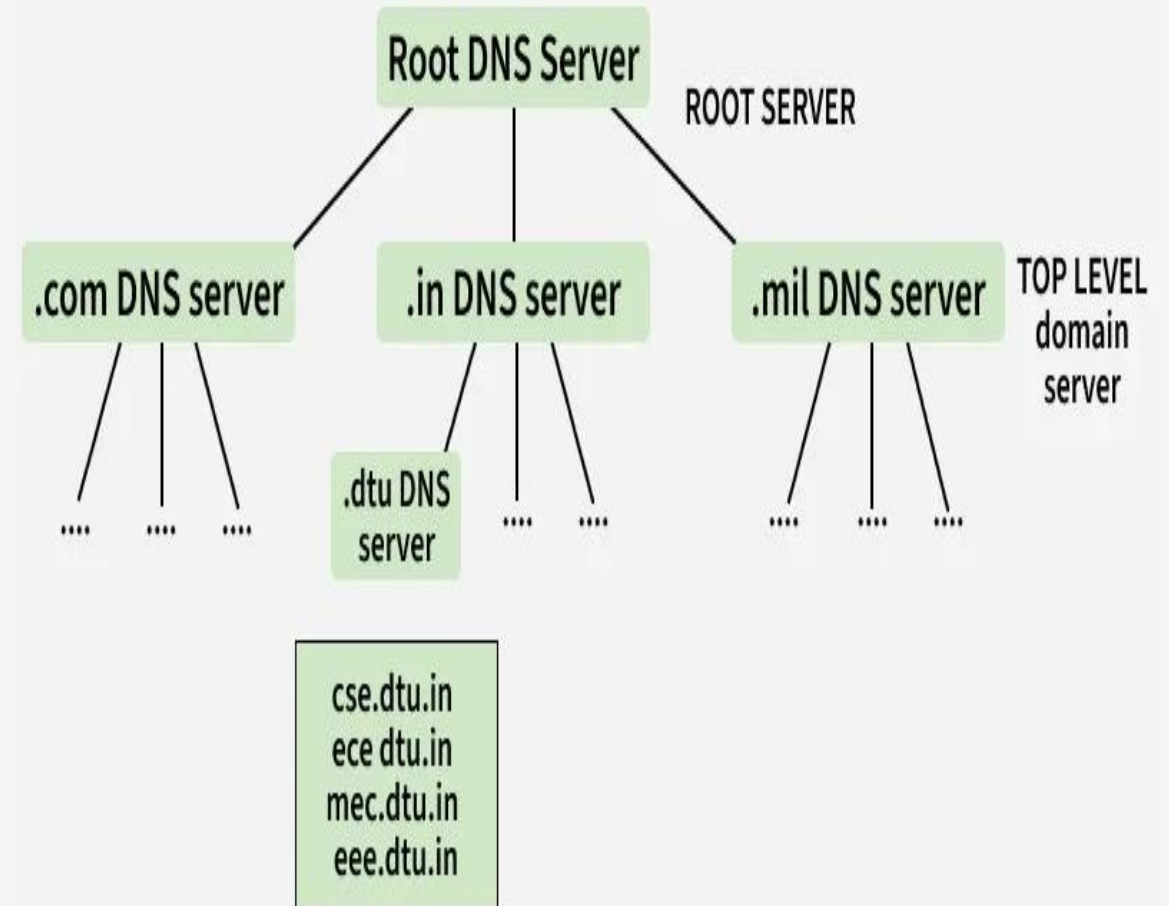
•**Final Response:** The DNS resolver sends the IP address to your computer, allowing it to connect to the website's server and load the page.

Structure of DNS

Root DNS Servers: These are the highest-level DNS servers and know where to find the TLD servers. They are crucial for directing DNS queries to the correct locations.

TLD Servers: These servers manage domain extensions like .com, .org, .net, .edu, .gov and others. They help route queries to the authoritative DNS servers for specific domains.

Authoritative DNS Servers: These are the servers that store the actual DNS records for domain names. They are responsible for providing the correct IP addresses that allow users to reach websites.



Domain Name Server

The client machine sends a request to the local name server, which, if the root does not find the address in its database, sends a request to the root name server, which in turn, will route the query to a top-level domain (TLD) or authoritative name server.

The root name server can also contain some hostName to IP address mappings.

The Top-level domain (TLD) server always knows who the authoritative name server is.

So finally the IP address is returned to the local name server which in turn returns the IP address to the host.

DNS Lookup

DNS Lookup, also called DNS Resolution, is the process of translating a human-readable domain name into its corresponding IP address. The process involves:

DNS Resolver: Initiates the lookup process. Also called a DNS client.

Recursive Query: Resolver queries multiple servers on behalf of the client until the IP is found.

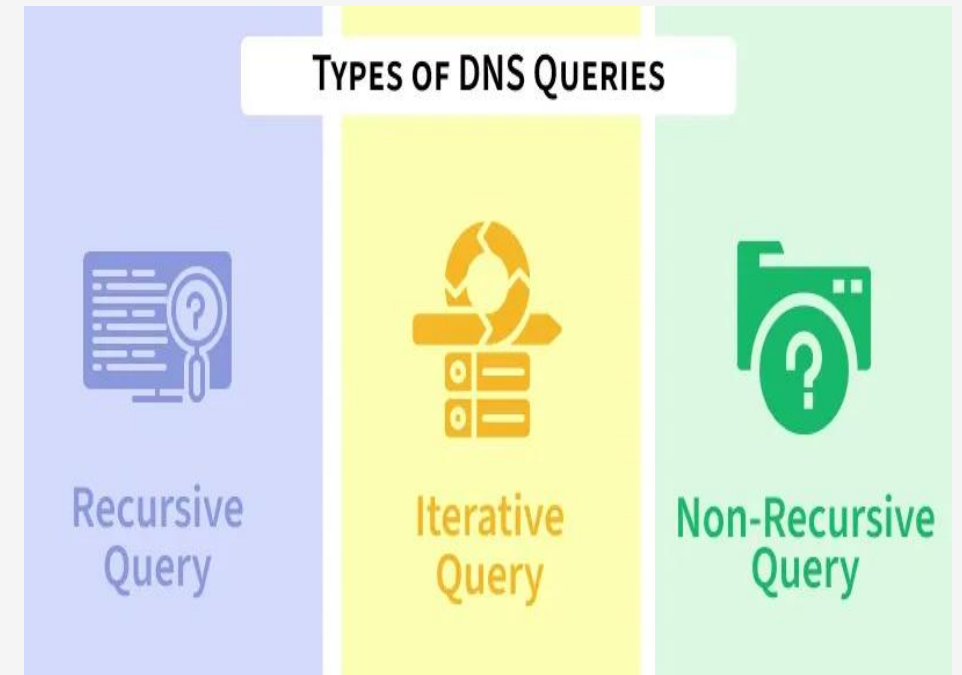
Iterative Query: Resolver asks servers for the best answer available.

Non-Recursive Query: Resolver queries a server that already has the record in its cache

Recursive Query: In this query, if the resolver is unable to find the record, in that case, DNS client wants the DNS Server will respond to the client in any way like with the requested source record or an error message.

Iterative Query: Iterative Query is the query in which DNS Client wants the best answer possible from the DNS Server.

Non-Recursive Query: Non-Recursive Query is the query that occurs when a DNS Resolver queries a DNS Server for some record that has access to it because of the record that exists in its cache.



DNS Caching and TTL (Time-to-Live)

DNS caching is a mechanism that stores DNS records locally to avoid querying external DNS servers repeatedly for the same information. This speeds up the browsing experience and reduces network traffic.

TTL (Time-to-Live) is the amount of time that a DNS record is cached before it expires. When the TTL expires, the cache is cleared and a fresh DNS query must be made. The TTL is defined by the authoritative DNS server, which can adjust this based on the nature of the domain.

DNS Security Extensions (DNSSEC)

It is a protocol designed to address these security concerns.

It adds cryptographic signatures to DNS records, allowing resolvers to verify the authenticity and integrity of DNS responses.

DNSSEC ensures that the information a user receives from a DNS query has not been tampered with.

Reverse DNS Lookup

Reverse DNS Lookup is the process of mapping an IP address back to its corresponding domain name. This is the opposite of the typical DNS lookup, where a domain name is resolved to an IP address. Reverse DNS is commonly used for:

Network Diagnostics: System administrators use reverse DNS to determine the domain name associated with a specific IP address. This helps identify the source of network traffic.

Email Security: Many email servers perform reverse DNS lookups to verify that incoming emails are coming from legitimate sources. This helps prevent spam and fraud.





Key Parts of the Web

•**URL (Uniform Resource Locator)**: This is the address of a webpage, like `https://www.example.com/`. It tells your browser exactly where to find the page.

•**HTTP (Hypertext Transfer Protocol)**: This is the set of rules that lets your browser and the server talk to each other to send and receive webpages.

•**HTML (Hypertext Markup Language)**: This is the code that tells browsers how to display a webpage, including where to put text, pictures, and links.

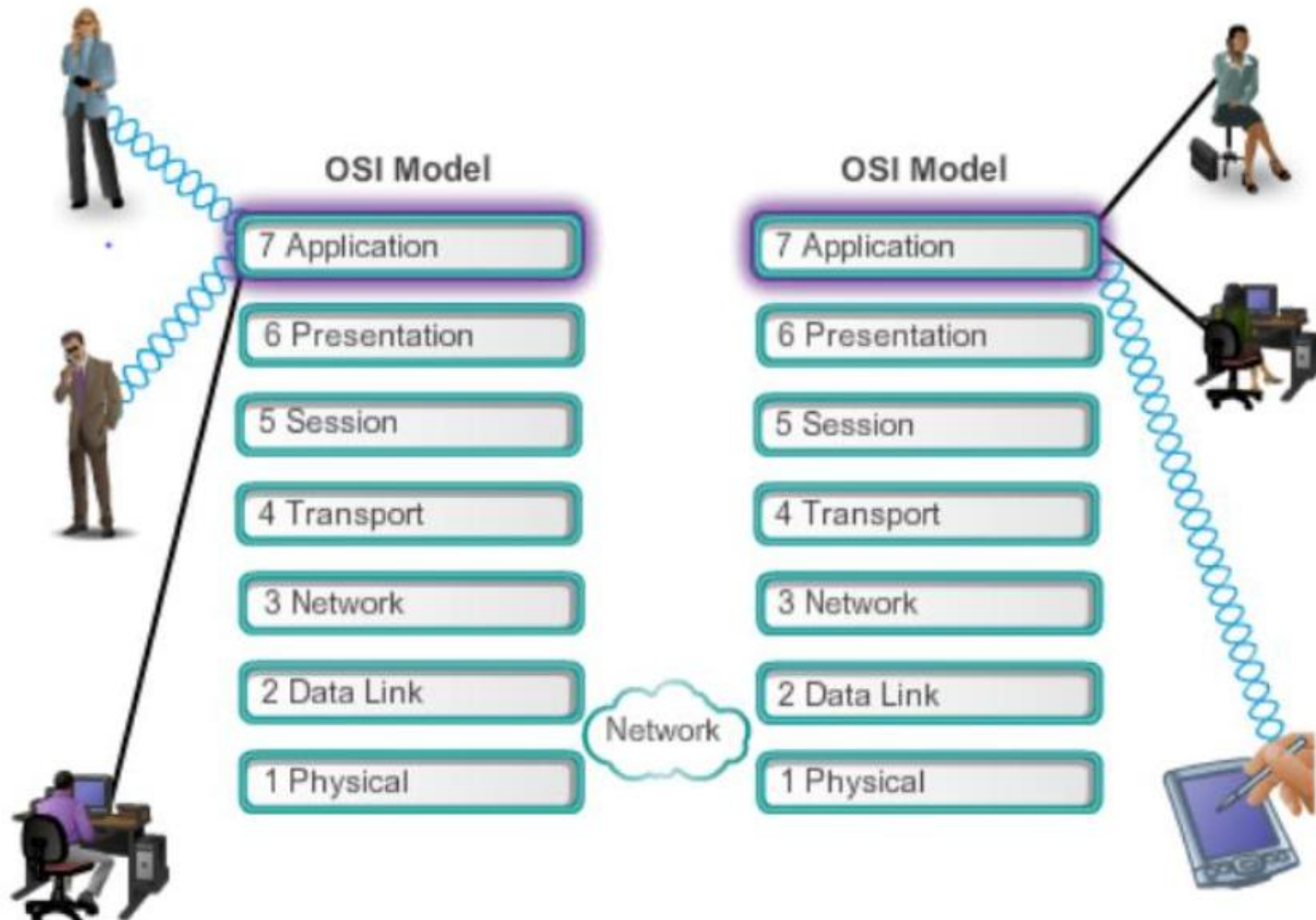
A Web browser is used to access web pages. Web browsers can be defined as programs which display text, data, pictures, animation and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers. Initially, Web browsers were used only for surfing the Web but now they have become more universal.

Key Parts of the Web

The below diagram indicates how the Web operates just like client-server architecture of the internet. When users request web pages or other information, then the web browser of your system request to the server for the information and then the web server provide requested services to web browser back and finally the requested service is utilized by the user who made the request.



The Application Layer provides the interface to the network.

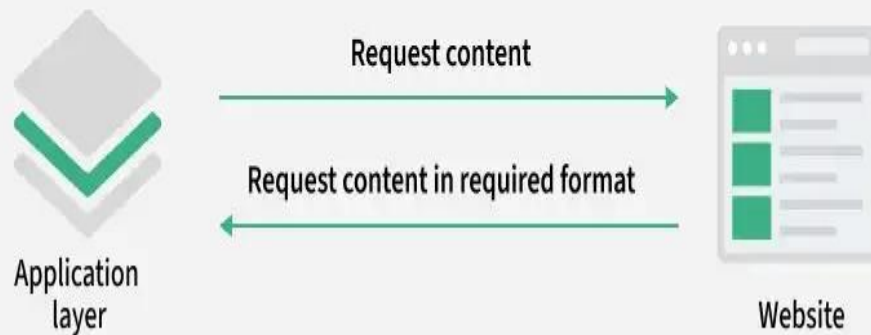


Functions of Application Layer

1. Data Representation
2. Network Service Access
3. Application Protocols
4. Session Management

1. Data Representation

It ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.



2. Network Service Access

It provides applications with access to network services, enabling communication over a network.



3. Application Protocols

It supports protocols like HTTP, FTP, SMTP, and DNS that enable communication between applications.



FTP



HTTP/S



SMTP



Telnet

4. Session Management

It manage establishment, maintenance, and termination of communication sessions between applications.



Working of Application Layer

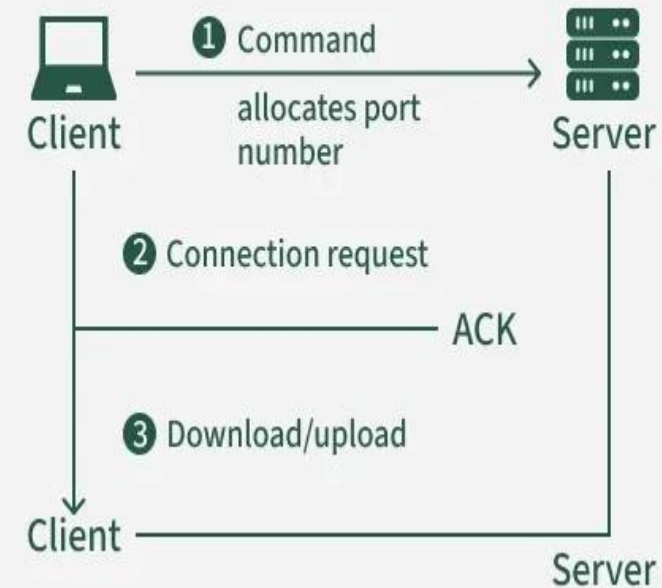
Once the data exchange is complete, the client or server can terminate the session and the connection is closed gracefully.

At first, client sends a command to server and when server receives that command, it allocates port number to client.

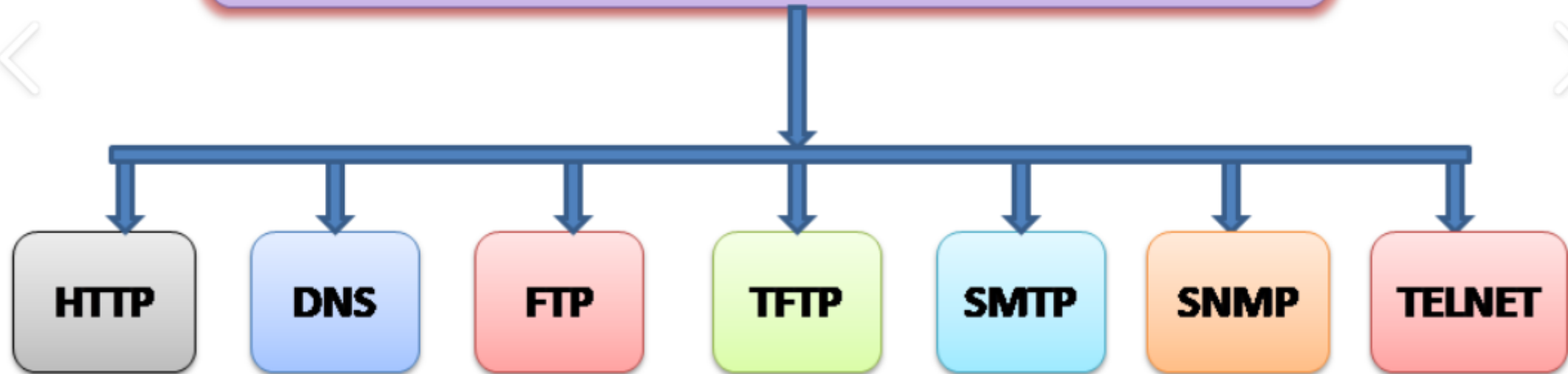
Thereafter, the client sends an initiation connection request to server and when server receives request, it gives acknowledgement (ACK) to client through which client has successfully established a connection with the server.

Therefore, now client has access to server through which it may either ask server to send any types of files or other documents or it may upload some files or documents on server itself.

WORKING OF APPLICATION LAYER



Protocols of Application Layer



Protocols in the Application Layer

HTTP (Hypertext Transfer Protocol): Used for transferring web content and hypermedia documents. It operates on port 80.

DNS (Domain Name System): Translates domain names into IP addresses, enabling access to internet resources. It uses port 53.

FTP (File Transfer Protocol): Facilitates file sharing and transfer between devices. It uses ports 20 (data) and 21 (control).

SMTP (Simple Mail Transfer Protocol): Handles email transmission between users. It operates on ports 25 and 587.

TELNET: Allows remote access to devices and systems for terminal communication. It uses port 23.

DHCP (Dynamic Host Configuration Protocol): Assigns IP addresses to devices dynamically. It uses ports 67 and 68.

TFTP: The **Trivial File Transfer Protocol (TFTP)** is a simple protocol designed for transferring files between devices in a network. It operates on the **application layer** of the TCP/IP model and uses **UDP (User Datagram Protocol)** as its transport layer, specifically on **port 69**.

Internet protocols

are a set of rules that allow computers and other devices to communicate over the Internet. These protocols ensure that data is sent, received, and understood correctly between different systems. There are many types of internet protocols, each serving a specific purpose, such as transferring files, sending emails, or securing data.

An [IP address](#) represents an Internet Protocol address. A unique address that identifies the device over the network. It is almost like a set of rules governing the structure of data sent over the Internet or through a local network. An IP address helps the Internet to distinguish between different [routers](#), computers, and websites.

Working of Internet Protocol

Dividing Data into Packets: When you send information over the internet, IP split it into small parts called packets. Each packet contains a piece of the data and the address of where it needs to go.

Addressing: Every device connected to the internet has its own IP address. This address helps identify where the data is being sent from and where it should be delivered.

Routing the Packets: As the packets travel across the internet, they pass through several devices called routers. These routers help direct the packets toward the correct destination, like how mail is sorted at different post offices.

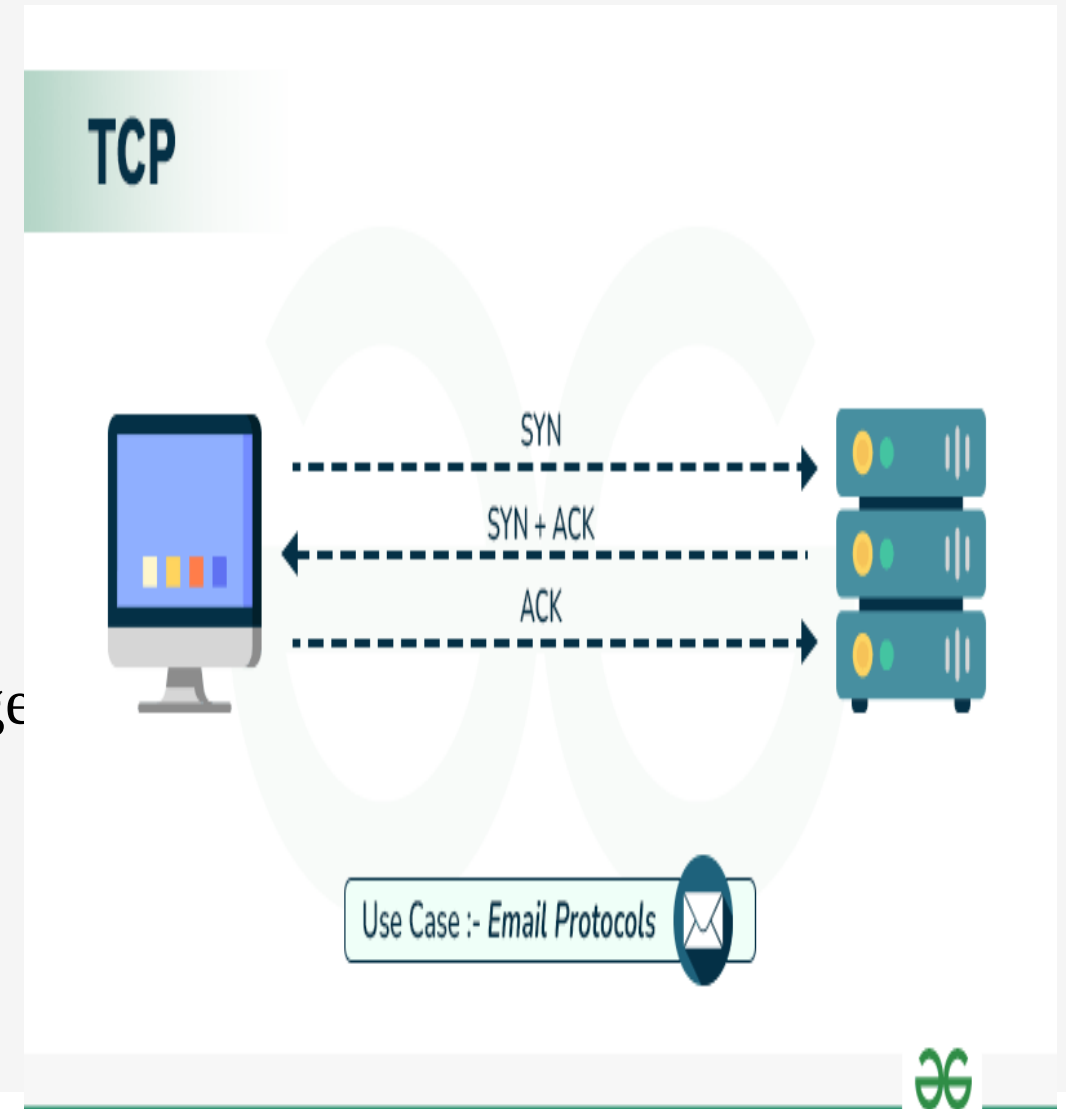
Reassemble the Data: Once all the packets arrive at the destination, they are put back together to recreate the original message or file.

Handling Missing Packets:

If some packets don't arrive, the system can request that they be sent again, making sure the complete data is received.

TCP/IP(Transmission Control Protocol/ Internet Protocol)

In TCP/IP, the IP protocol ensures that each computer that is connected to the Internet is having a specific serial number called the IP address. TCP specifies how data is exchanged over the internet and how it should be broken into IP packets. It also makes sure that the packets have information about the source of the message data, the destination of the message data, the sequence in which the message data should be re-assembled, and checks if the message has been sent correctly to the specific destination. The TCP is also known as a connection-oriented protocol.

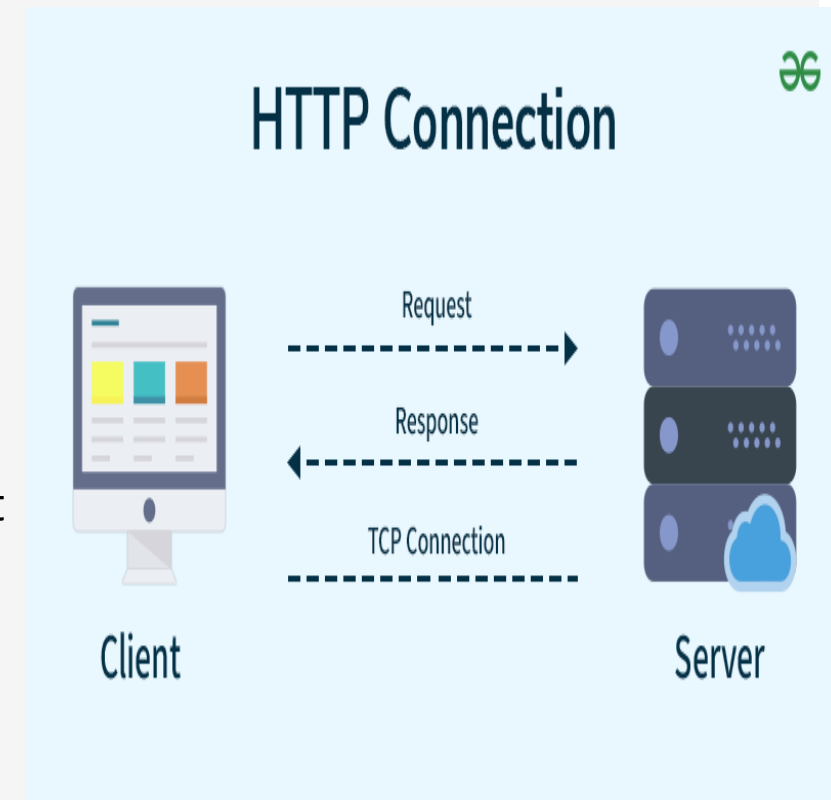


SFTP(Secure File Transfer Protocol) & HTTP(Hyper Text Transfer Protocol)

SFTP which is also known as SSH FTP refers to File Transfer Protocol (FTP) over Secure Shell (SSH) as it encrypts both commands and data while in transmission. SFTP acts as an extension to SSH and encrypts files and data then sends them over a secure shell data stream. This protocol is used to remotely connect to other systems while executing commands from the command line.

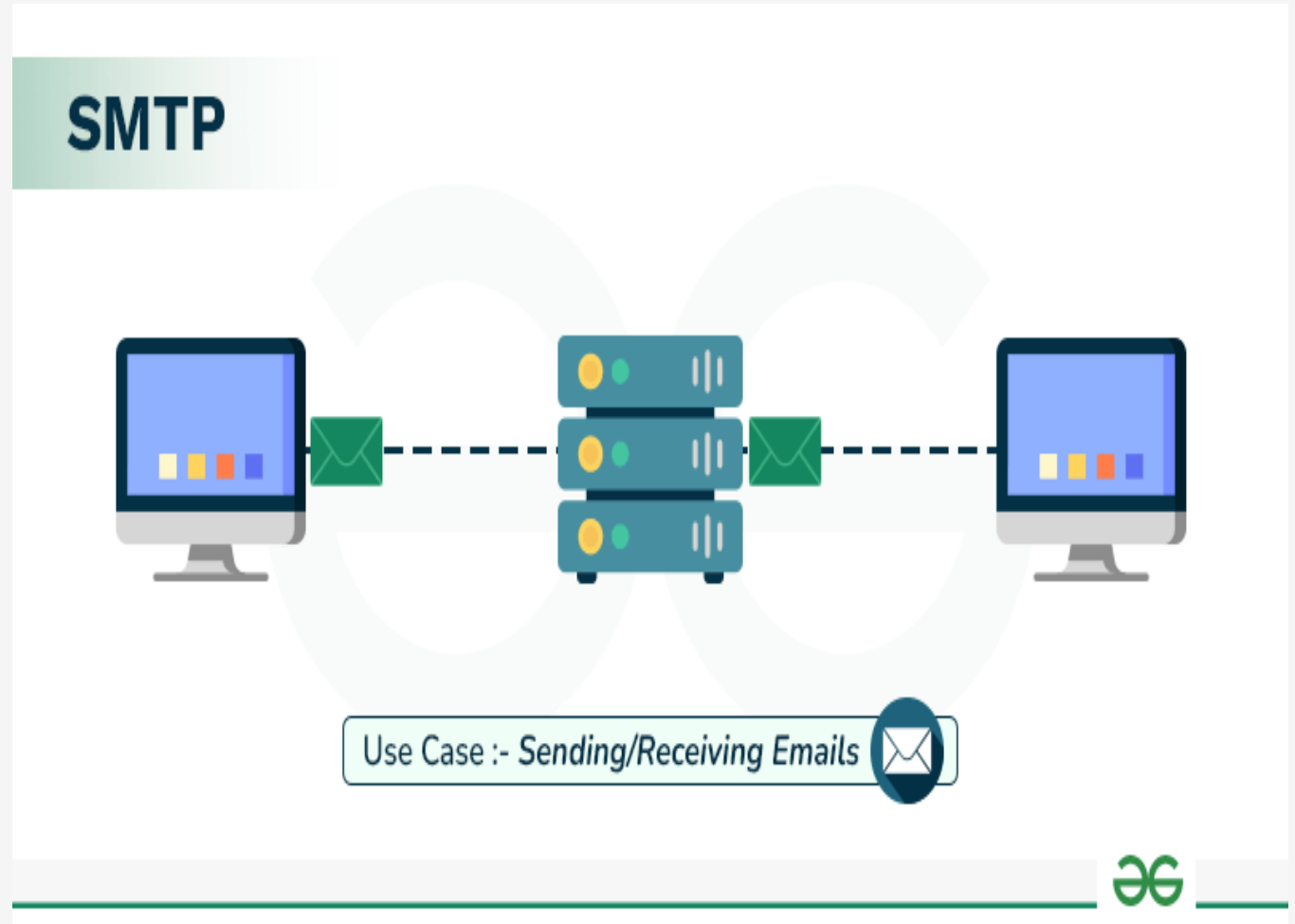
HTTP(Hyper Text Transfer Protocol)

HTTP protocol is used to transfer hypertexts over the internet and it is defined by the www(world wide web) for information transfer. This protocol defines how the information needs to be formatted and transmitted. And, it also defines the various actions the web browsers should take in response to the calls made to access a particular web page. Whenever a user opens their web browser, the user will indirectly use HTTP as this is the protocol that is being used to share text, images, and other multimedia files on the World Wide Web.



SMTP(Simple Mail Transfer Protocol)

SMTP protocol is important for sending and distributing outgoing emails. This protocol uses the header of the mail to get the email id of the receiver and enters the mail into the queue of outgoing mail. And as soon as it delivers the mail to the receiving email id, it removes the email from the outgoing list. The message or the electronic mail may consider the text, video, image, etc. It helps in setting up some communication server rules.

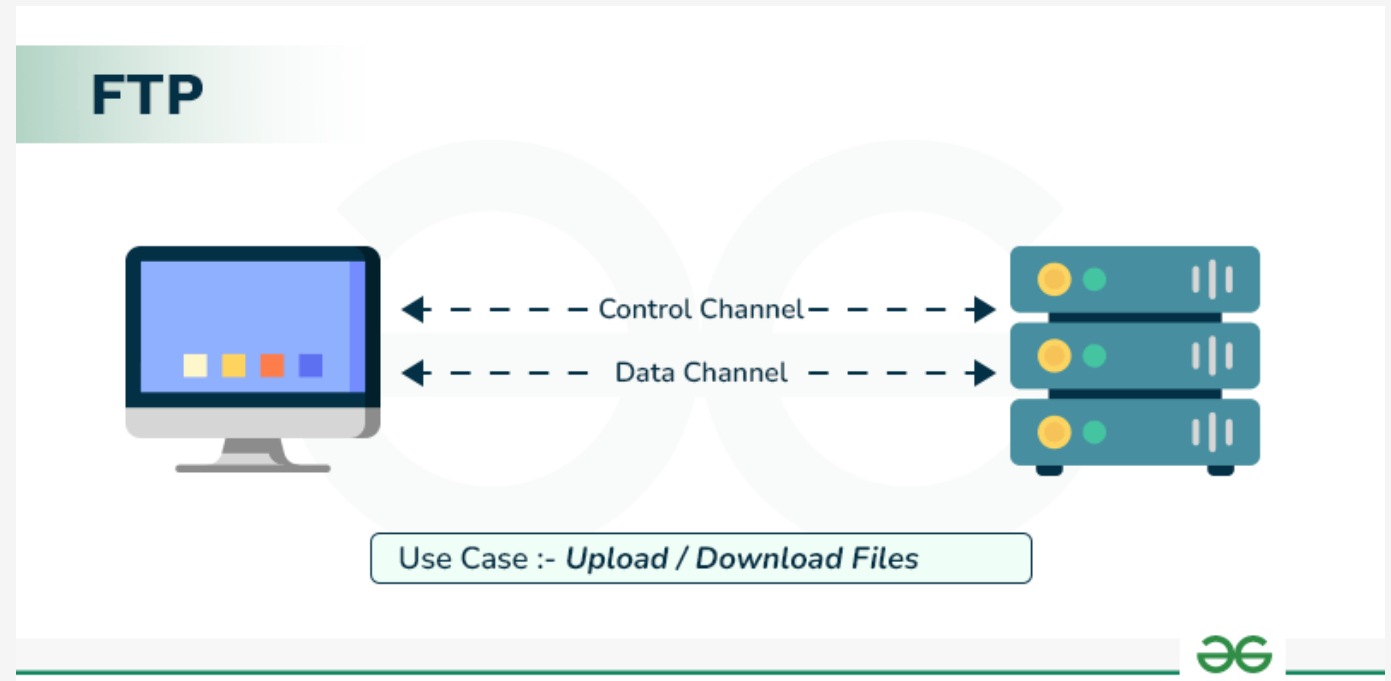


PPP(Point-to-Point Protocol) & FTP (File Transfer Protocol)

PPP is a communication protocol that is used to create a direct connection between two communicating devices. This protocol defines the rules using which two devices will authenticate with each other and exchange information with each other. For example, A user connects his PC to the server of an Internet Service Provider and also uses PPP. Similarly, for connecting two routers for direct communication it uses PPP.

FTP (File Transfer Protocol)

This protocol is used for transferring files from one system to the other. This works on a client-server model. When a machine requests for file transfer from another machine, the FTO sets up a connection between the two and authenticates each other using their ID and Password. And, the desired file transfer takes place between the machines.

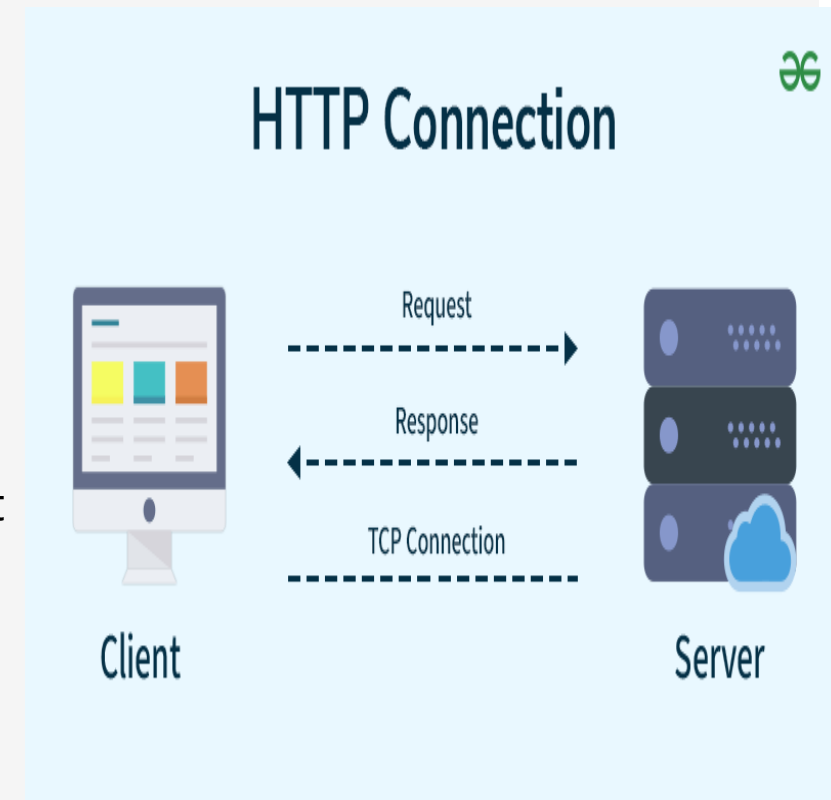


SFTP(Secure File Transfer Protocol) & HTTP(Hyper Text Transfer Protocol)

SFTP which is also known as SSH FTP refers to File Transfer Protocol (FTP) over Secure Shell (SSH) as it encrypts both commands and data while in transmission. SFTP acts as an extension to SSH and encrypts files and data then sends them over a secure shell data stream. This protocol is used to remotely connect to other systems while executing commands from the command line.

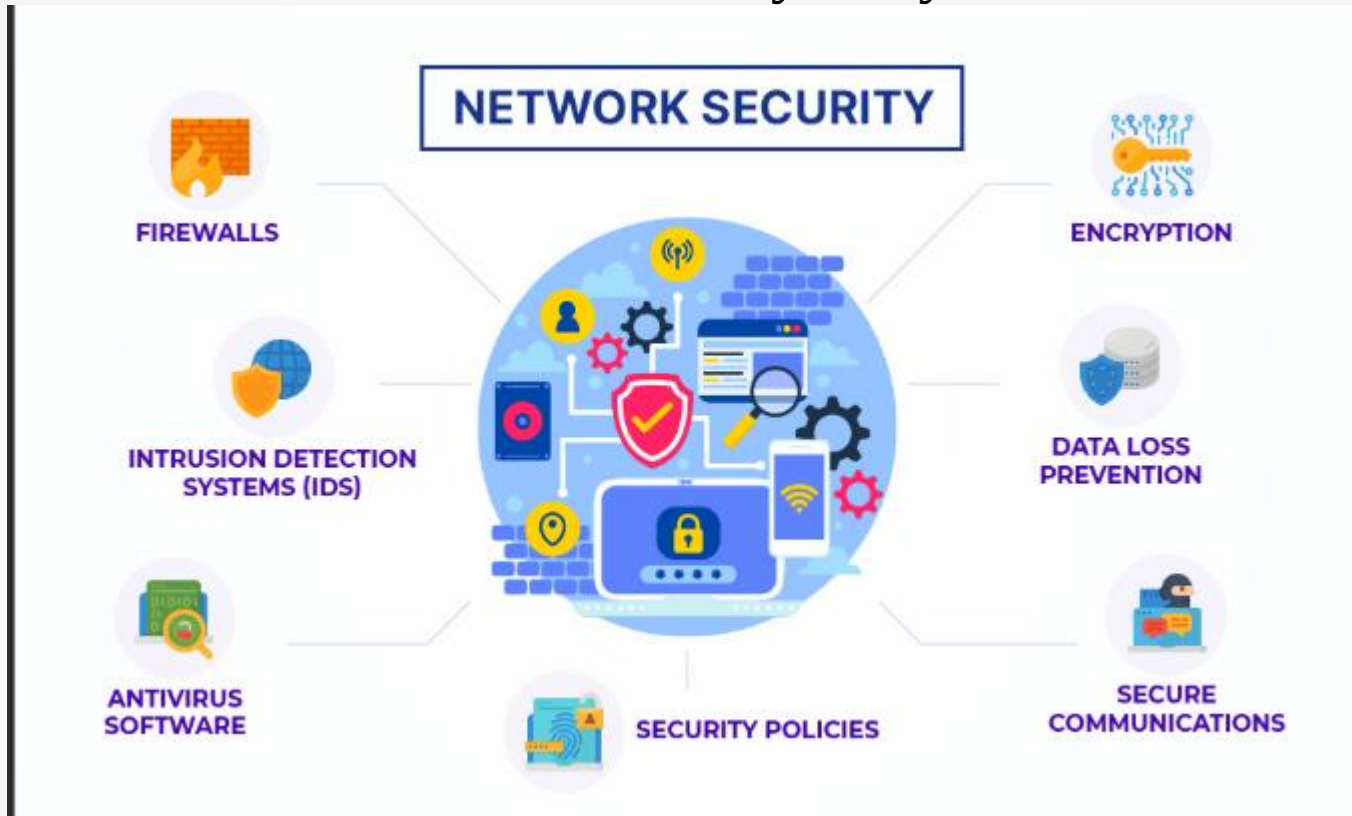
HTTP(Hyper Text Transfer Protocol)

HTTP protocol is used to transfer hypertexts over the internet and it is defined by the www(world wide web) for information transfer. This protocol defines how the information needs to be formatted and transmitted. And, it also defines the various actions the web browsers should take in response to the calls made to access a particular web page. Whenever a user opens their web browser, the user will indirectly use HTTP as this is the protocol that is being used to share text, images, and other multimedia files on the World Wide Web.

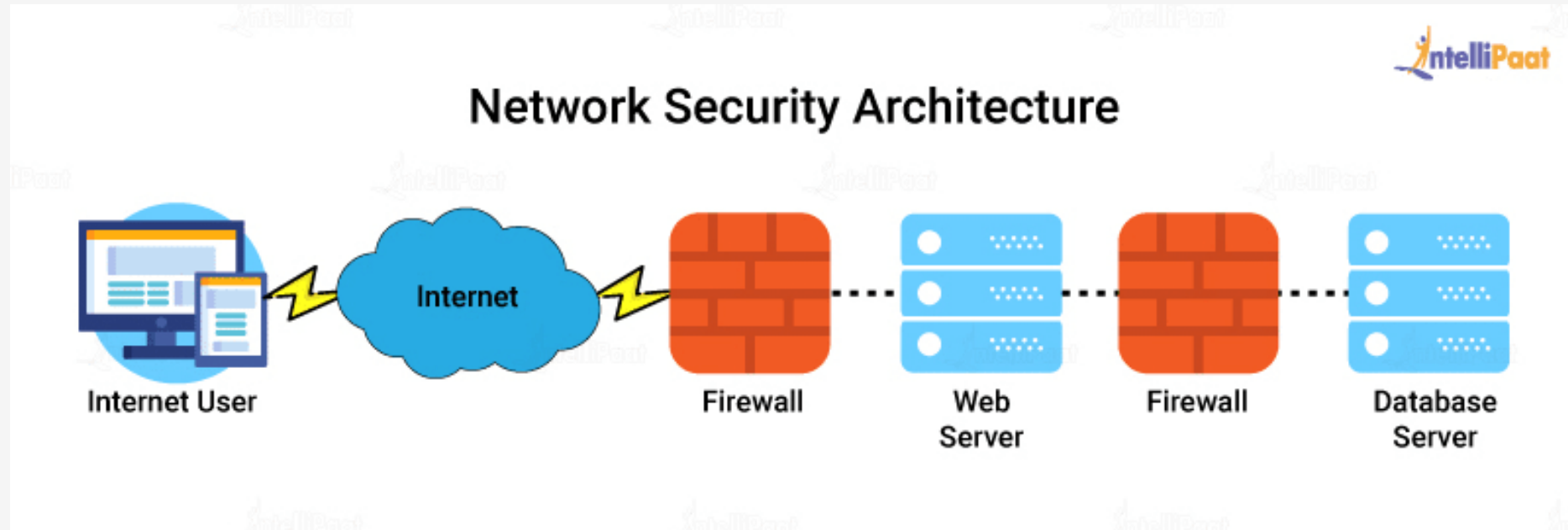


Network security

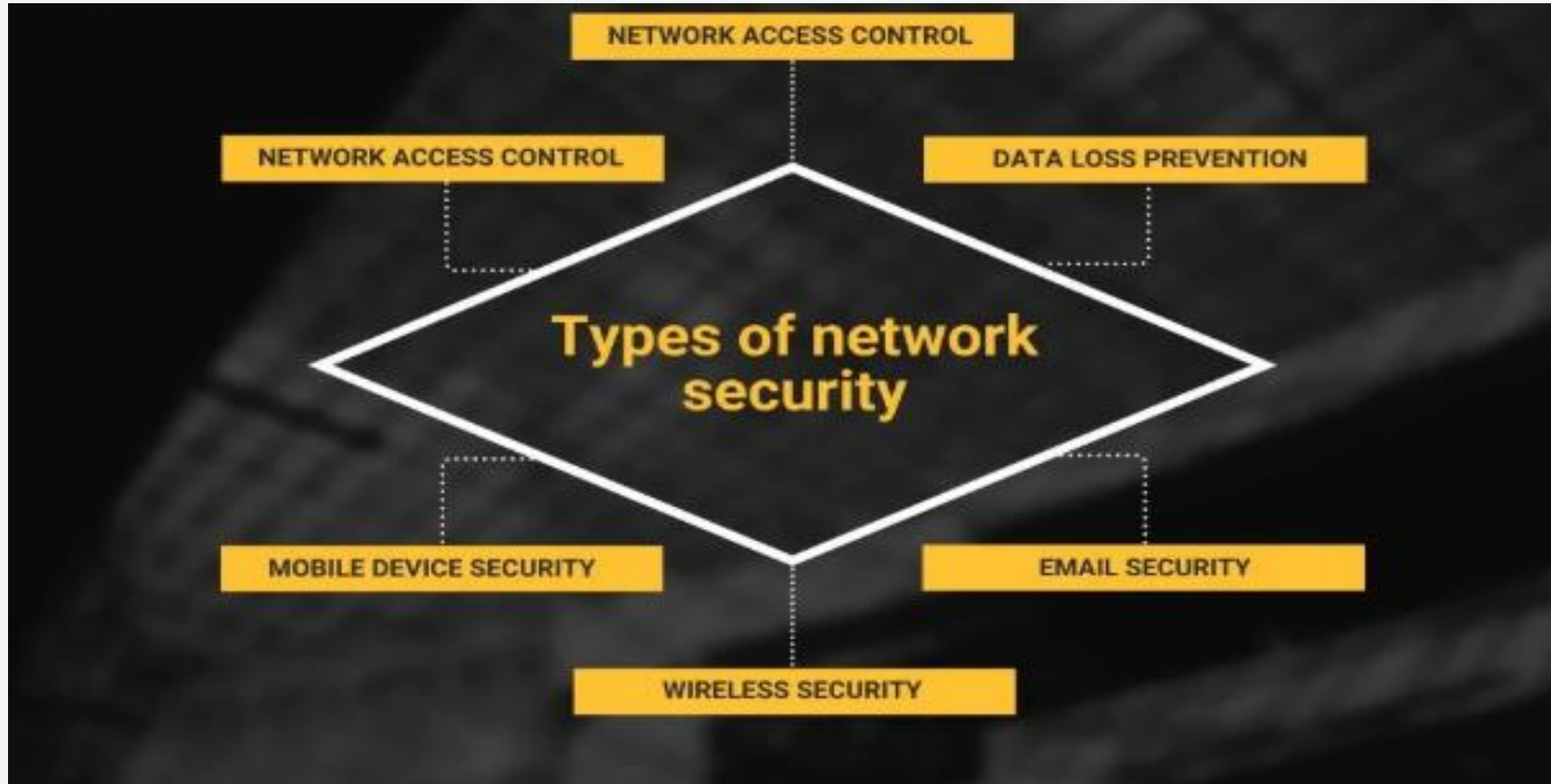
encompasses the practices and technologies designed to protect the integrity, confidentiality, and availability of computer networks and the data they carry.



Network security



Network security



Physical Network Security: It is the foundational level of network protection, focused on preventing unauthorized individuals from gaining physical access to networking equipment and compromising the confidentiality of the network. This can be achieved through measures such as biometric systems, access cards, and other physical security devices.

Technical Network Security: It primarily focuses on protecting the data stored in the network or data involved in transitions through the network. This type serves two purposes. One is protected from unauthorized users, and the other is protected from malicious activities.

Administrative Network Security: This level of network security protects user behavior like how the permission has been granted and how the authorization process takes place. This also ensures the level of sophistication the network might need for protecting it through all the attacks. This level also suggests necessary amendments that have to be done to the infrastructure.

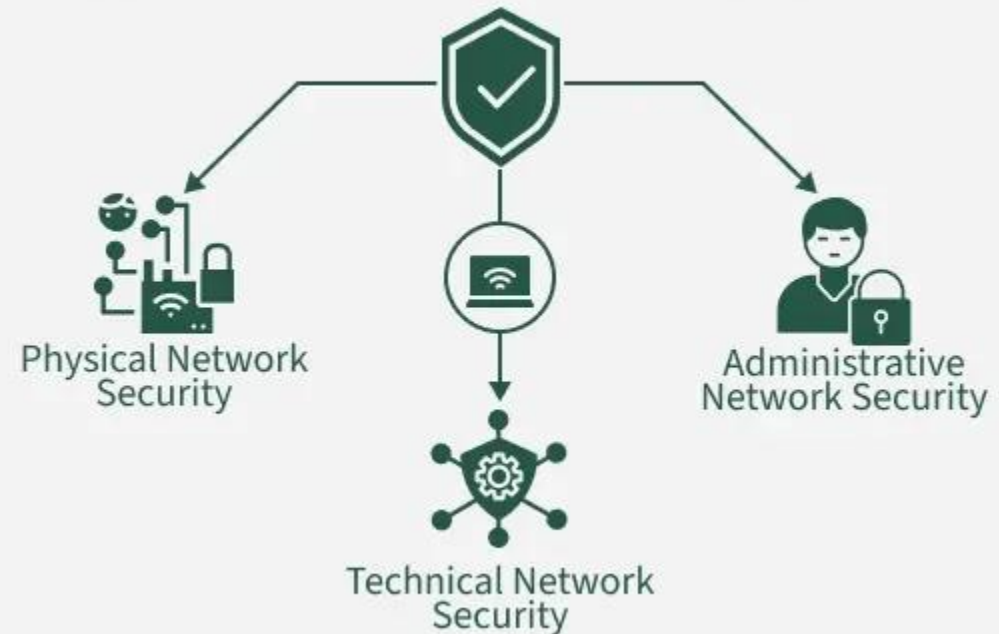
Firewall Security

Firewalls act as a barrier between trusted and untrusted networks by monitoring and filtering incoming and outgoing traffic based on predefined rules. They play a critical role in preventing unauthorized access while permitting legitimate communication.

Network security uses multiple layers of protection, both at the network edge and within the internal environment. Each layer enforces specific rules and controls to regulate access to network resources. Authorized users can safely utilize the network, while malicious actors and threats are blocked from causing harm.

The basic principle of network security is protecting huge stored data and networks in layers that ensure the enforcement of rules and regulations that have to be acknowledged before performing any activity on the data. These levels are:

HOW DOES NETWORK SECURITY WORK?

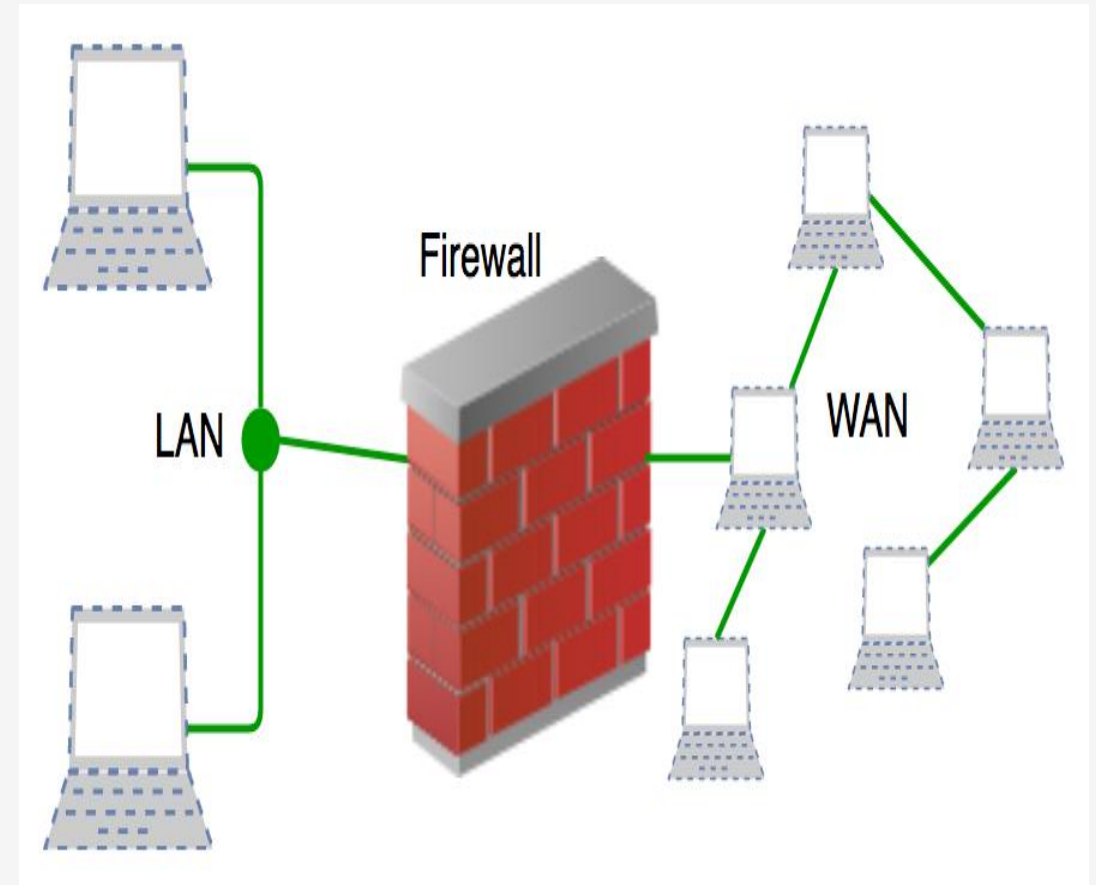


A firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either:

Accept: Allow the traffic.

Reject: Block with an error response.

Drop: Block silently without response.



Firewalls are essential because they:

Prevent Unauthorized Access: Like a locked door with a guard, only trusted users and traffic are allowed through.

Block Malicious Traffic: Harmful data such as viruses, phishing attempts, or denial-of-service (DoS) attacks are stopped before reaching the system.

Protect Sensitive Information: Safeguards personal and business data from theft or accidental leaks.

Control Network Usage: Enforces policies such as parental controls, workplace restrictions, or government filtering.

Mitigate Insider Risks: Detects suspicious applications or data exfiltration attempts from within the network.

Working of Firewall

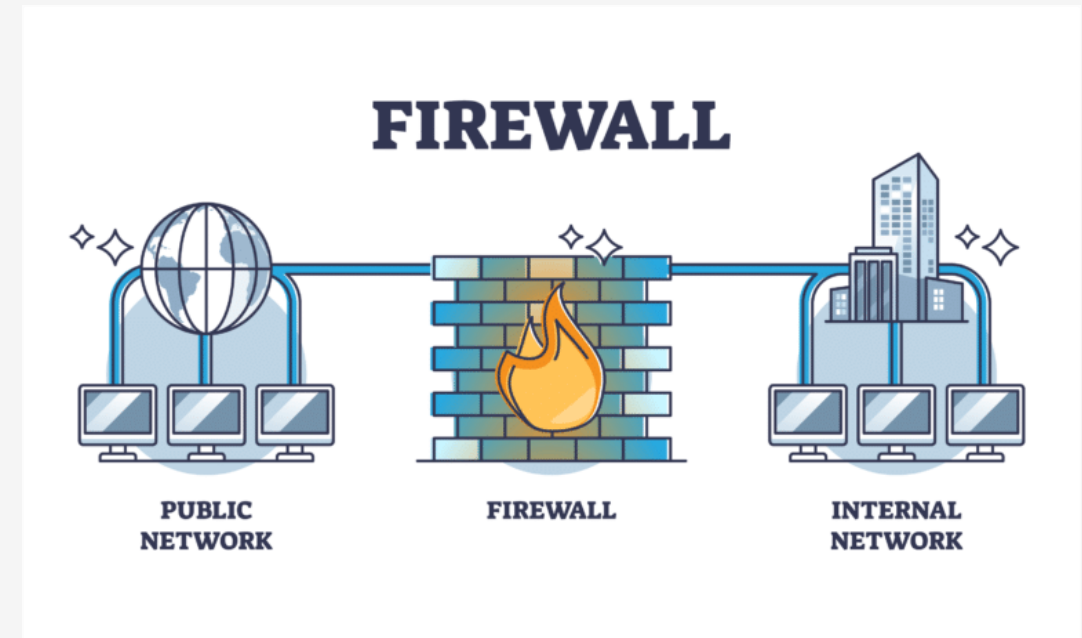
A firewall inspects all incoming and outgoing traffic and decide whether to allow or block it.

All data packets entering or leaving the network must first pass through the firewall. The firewall examines each packet against predefined security rules set by the organization.

If the packet matches safe rules, it is allowed; if it is suspicious, blacklisted, or contains malicious content, it is blocked.

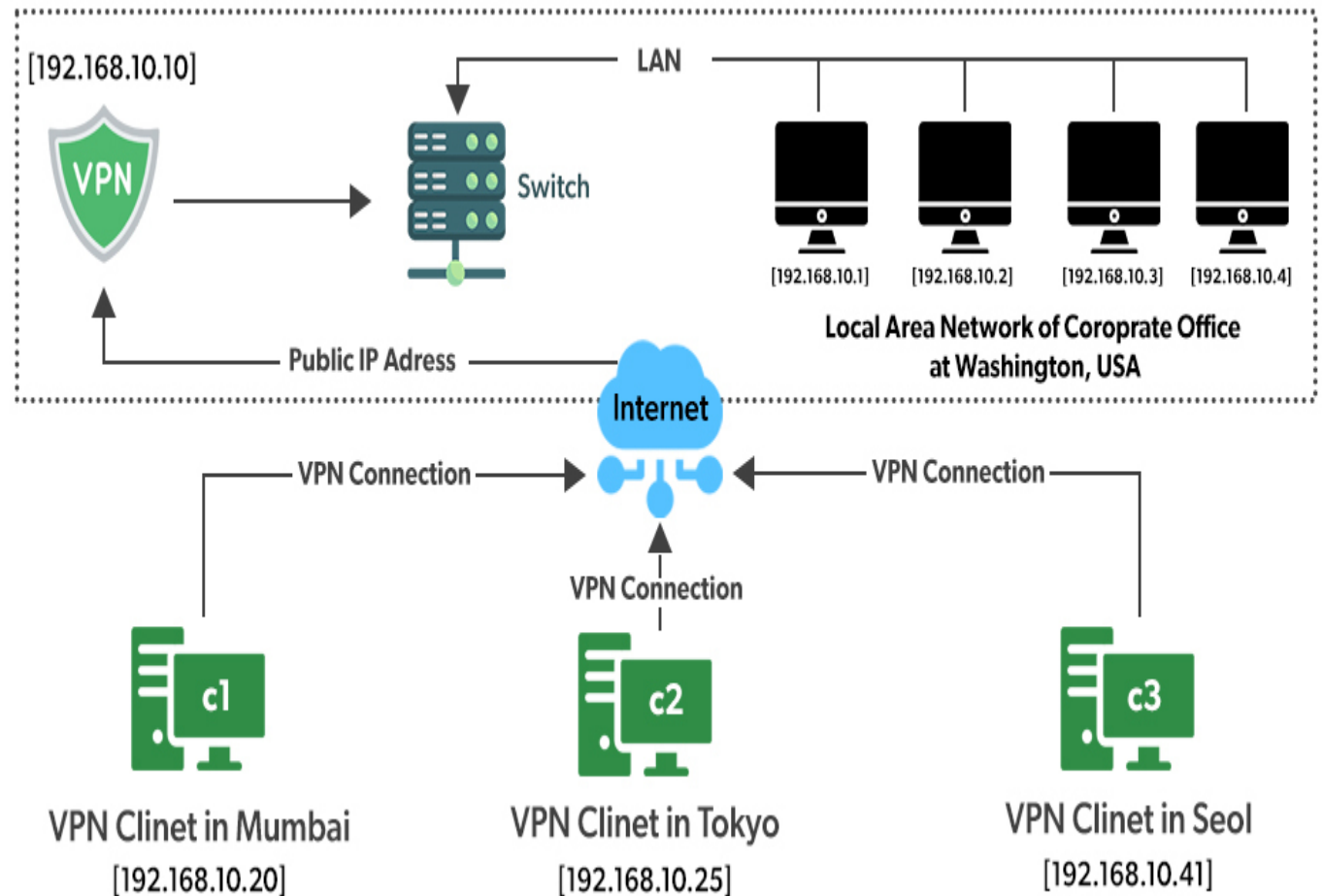
Blocked or unusual traffic is recorded in logs, and real-time alerts may be generated for serious threats.

Since it is not possible to define every rule, the firewall applies a default policy (accept, reject, or drop). Setting the default policy to drop or reject is considered best practice to prevent unauthorized access.



A VPN (Virtual Private Network)

A VPN (Virtual Private Network) is a technology that creates a secure, encrypted connection between your device and the internet. It acts as a private tunnel for your internet traffic, preventing unauthorized access and ensuring your online activities remain private. VPNs are essential for enhancing online privacy, protecting sensitive data, and enabling secure access to the internet, especially on public networks.





Types of VPNs

Remote Access VPN: This type allows users to connect to a private network from a remote location. It is commonly used by employees to access company resources securely while away from the office.

Site-to-Site VPN: Often used by large organizations, this VPN connects multiple networks, allowing different office locations to communicate securely.

Mobile VPN: Designed for mobile users, this VPN enables secure connections to private networks through cellular networks, ensuring data protection while on the go.

SSL VPN: This type uses the SSL protocol to secure connections, allowing remote users to access private networks through a web browser without needing additional software.

Core Role: The VPN Concentrator/Router

The most critical device is the VPN Gateway (or VPN Concentrator/Router). Its role is to:

Authentication: Authenticate the remote user/device attempting to connect.

Encryption and Decryption: Establish a secure tunnel using protocols like IPsec or SSL/TLS. All data entering the tunnel is encrypted (scrambled), and all data leaving the tunnel is decrypted.

Encapsulation: Take the original IP packet and wrap it inside a new, secure outer packet, effectively hiding the source and destination details.

Routing: Once the data is decrypted, the VPN gateway acts as a router to forward the packet onto the private internal network.

Supporting Device Roles

Firewall: The firewall is placed in front of (or sometimes integrated with) the VPN Gateway. Its primary roles are:

Policy Enforcement: Control which external hosts are even allowed to communicate with the VPN Gateway port.

Stateful Inspection: Track the VPN session to ensure only valid, established VPN traffic is allowed into the private network.

Standard Router: Standard routers on the internal network handle the traffic after it exits the VPN gateway. They ensure the decrypted traffic is routed efficiently to the correct internal server or client based on its private IP address.

Authentication Server (RADIUS/TACACS+): Often, the VPN gateway doesn't store user credentials but relies on an external authentication server to centrally verify the remote user's identity and permissions before allowing the connection to be established.

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